Holly AOI Training guideline

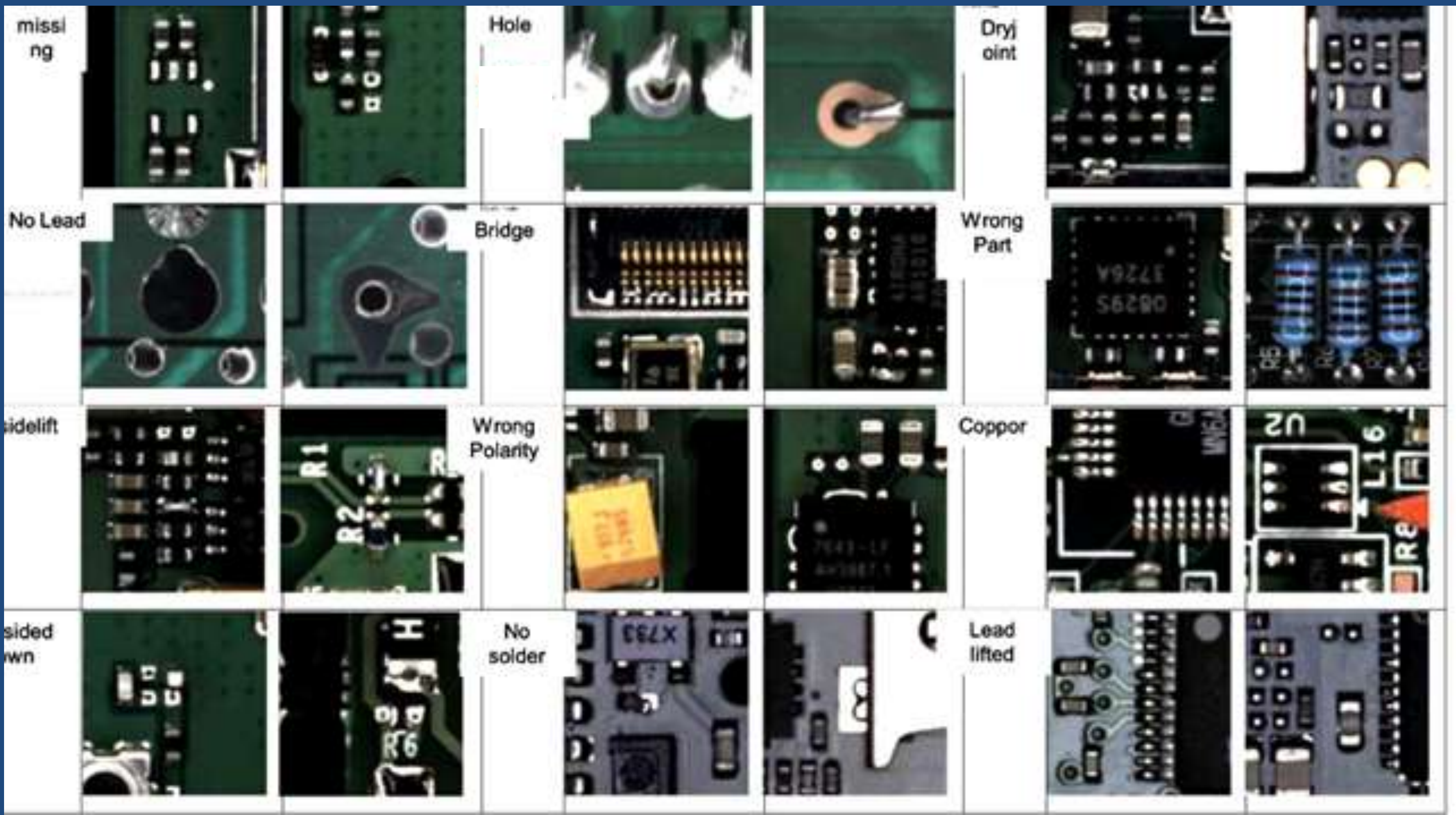
What is A O I?

A O I named Automated Optical Inspection System, it is used to inspect the status of appearance defects , in the SMT(surface mounted technology) field, it is hardly to check component s by human eyes as the mounted components become smaller , AOI will give a total solution to do check component s instead of human eyes

Defect name & Images introduction

What can AOI machine do check?

It can check for missing、shift、billboard、tombstone、flip over、polarity、wrong part、broken component、bridge=short circuit、dry-joint、no solder、insufficient solder、excess solder、copper、lift-chip、IC , lift-lead、XY offset output ,and so on



Main Part of Holly AOI's image system

Holly perfect Lighting System

Color digital camera

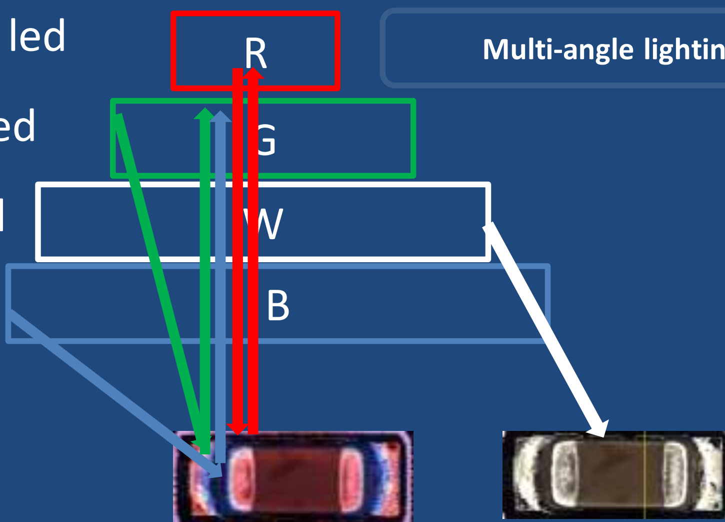
High quality telecentric lens

Red led

Green led

White led

Blue led

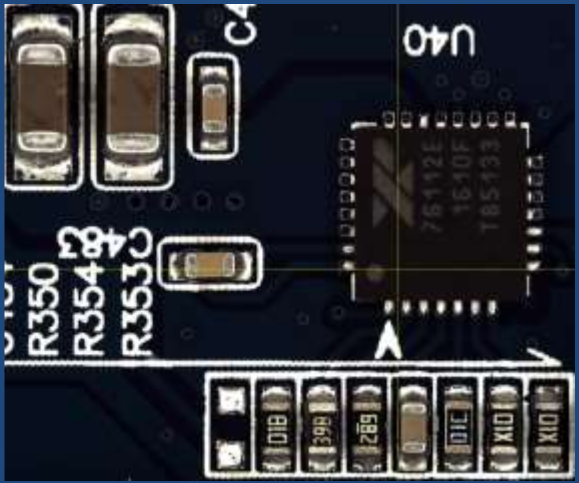


Multi-angle lighting

TOP SIDE

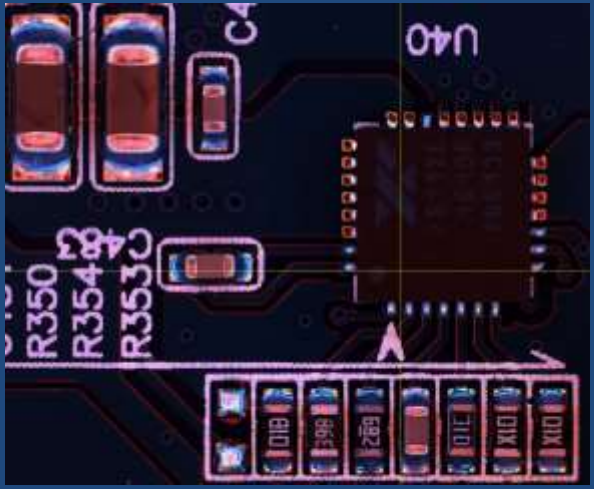
Holly define R&G&B as a group named "Top" image and W named "Side" image

The captured image effect by side



The angle lighting with the white LED can be used to show the real color of the objects which overcome the shortage of RGB TYPES aoi that is difficult to inspect the defect of no sold for copper pad

The captured image effect by top



It looks like color image with solder part, and RGB makes the solder parameter be predictable

TOP introduction

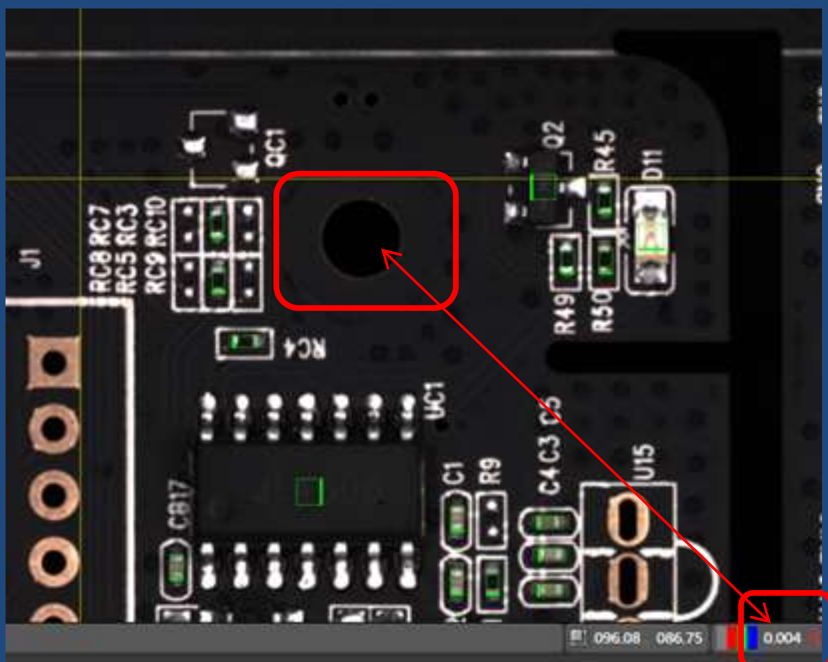
When the RGB light reflect to the flat surface on the object , it will reflect light to the camera directly , the captured image appear red color effect

When the RGB light reflect to the slope surface on the object ,the captured image appear green or blue effect

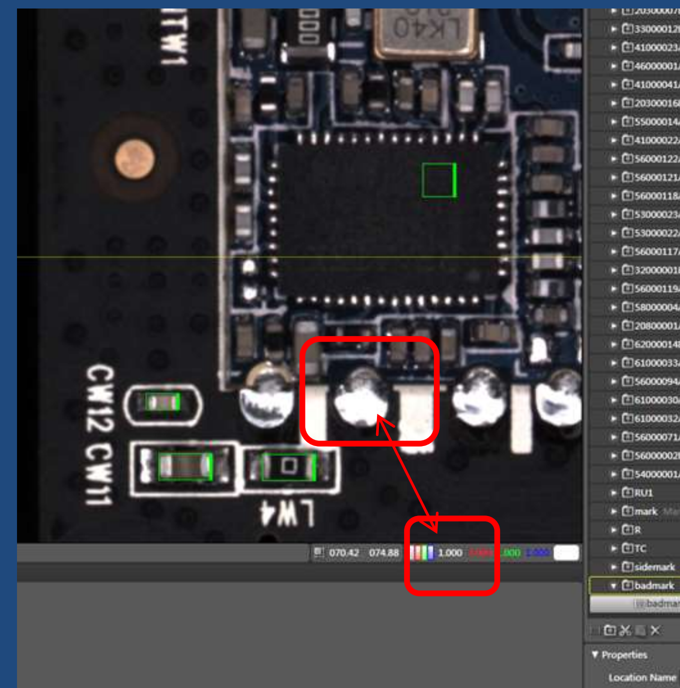
Luminance introduction

What is Luminance value of HOLLY software?

Definition: PCB entire image is made up of pixels,each pixel has a value , we name Luminance value, the maximum value is 1 and the minmum value is zero, in other words ,the blackest pixel value is 0 and the brightest pixel value is 1



Minimum value



Maxmum value

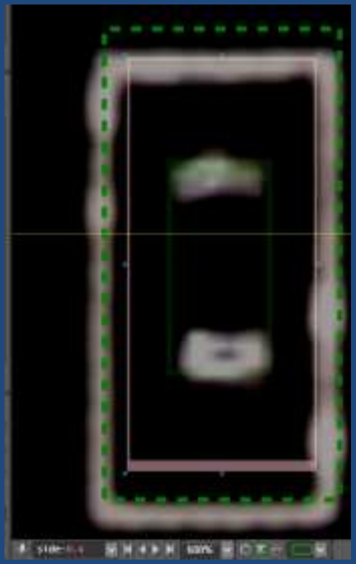
The method of adjust lighting source

Due to Interference by printed area or PCB board area , the distribution of computed result histogram is irregular, we need to adjust light source light source

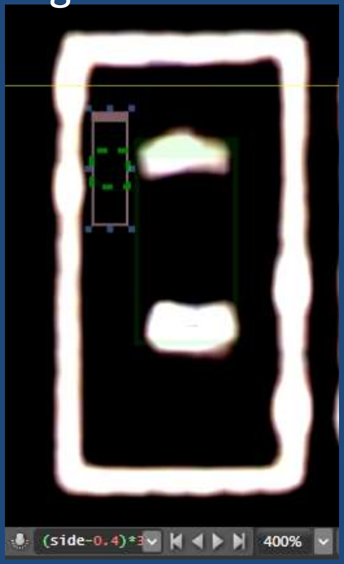
irregular

regular

IHL: Side



IHL: Side-0.4



IHL: (Side-0.4)*3

Computed process

Solder luminance value: 0.340
Electrode luminance value: 0.934

Solder luminance value: 0.340-0.4=-0.06=0
Electrode luminance value: 0.934-0.4=0.534

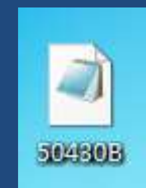
Solder luminance value: 0*3=0
Electrode luminance value: 0.534*3=1.6

the luminance value of each pixel in fact minus 0.4 and times 3 (we often adjust light source by minus a number which is less then 1)so that the luminance value around the inspect window equal to 0 ,and then multiply light source(packing the light source) by a number which is more than 1, if the computed result of the value is more than 1, it default 1, and is less than 0, it default 0

Cad file Introduction

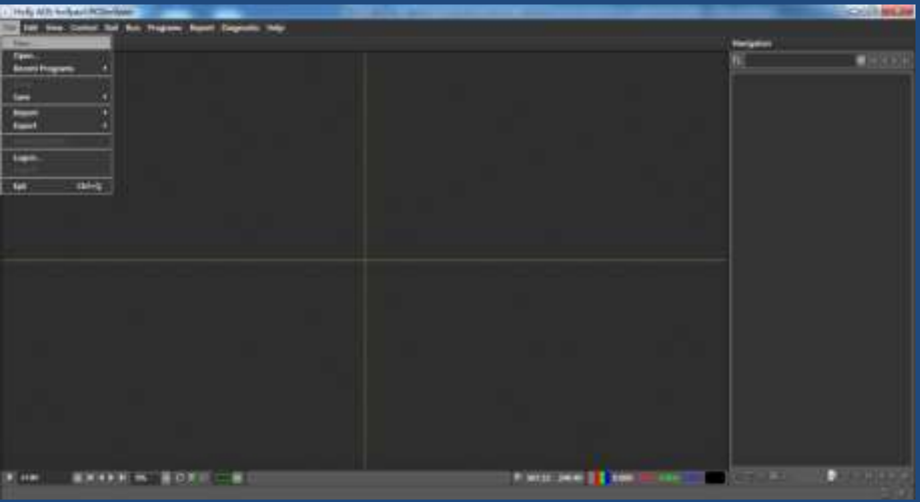
Before making programing, we need to get a cad document which including location, X coordinate , Y coordinate , angle , and part number and so on. And its format is “txt”

```
Unit: mm
Board Name: 50430B
Board Width: 170
Board Height: 97
Total Component Number: 1204
No., Location Name, Block, Catalog, X, Y, Rotate
1, C0503, 1, 540000001587, 14.7813, 15.7269, 180
2, C0553, 1, 540000001587, 19.8376, 14.7394, 0
3, C0503, 2, 540000001587, 38.2866, 83.5354, 0
4, C0553, 2, 540000001587, 33.1929, 84.5041, 180
5, C0503, 3, 540000001587, 44.8626, 15.8894, 180
6, C0503, 5, 540000001587, 74.7376, 15.7394, 180
```

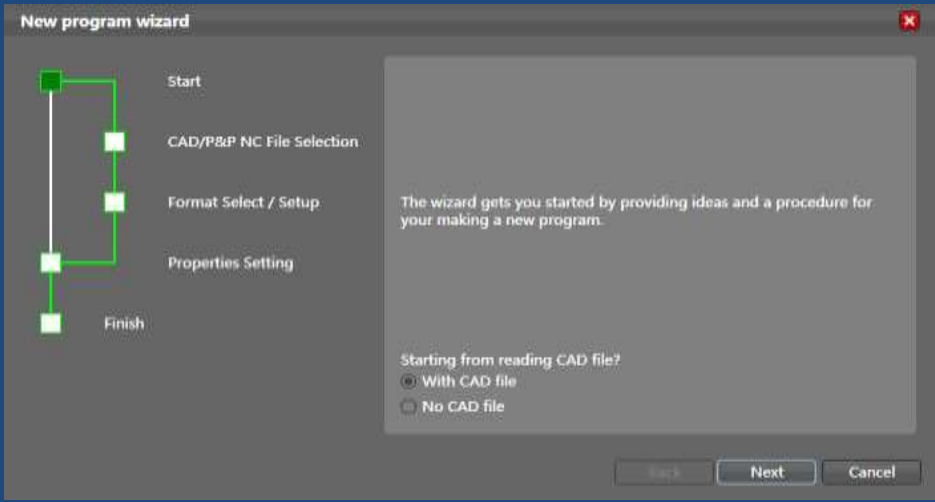


Importing CAD file method Introduction

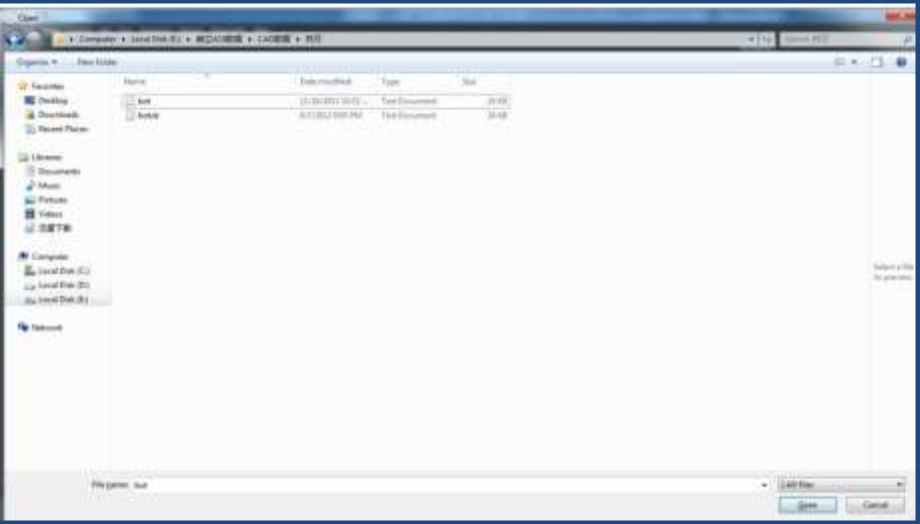
1.Click “file” menu, select “new…” submenu items；



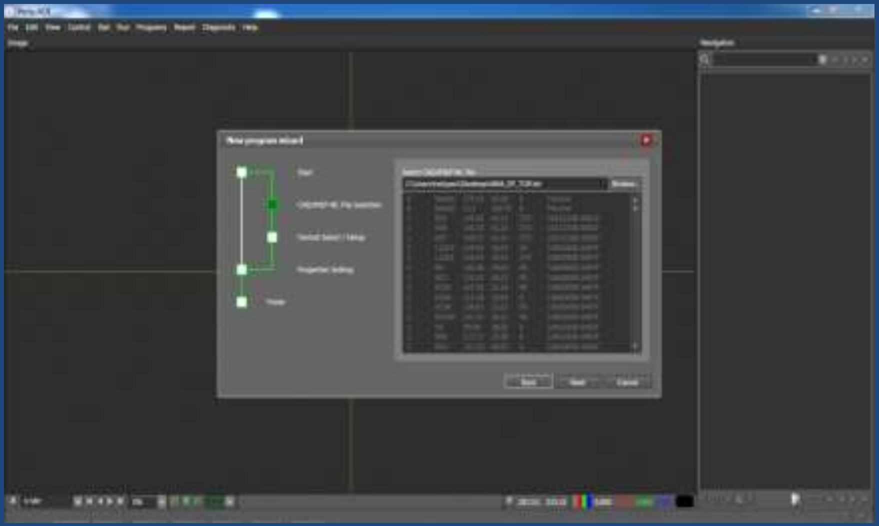
2. to select “with CAD file” radio button， click “next” button to continue



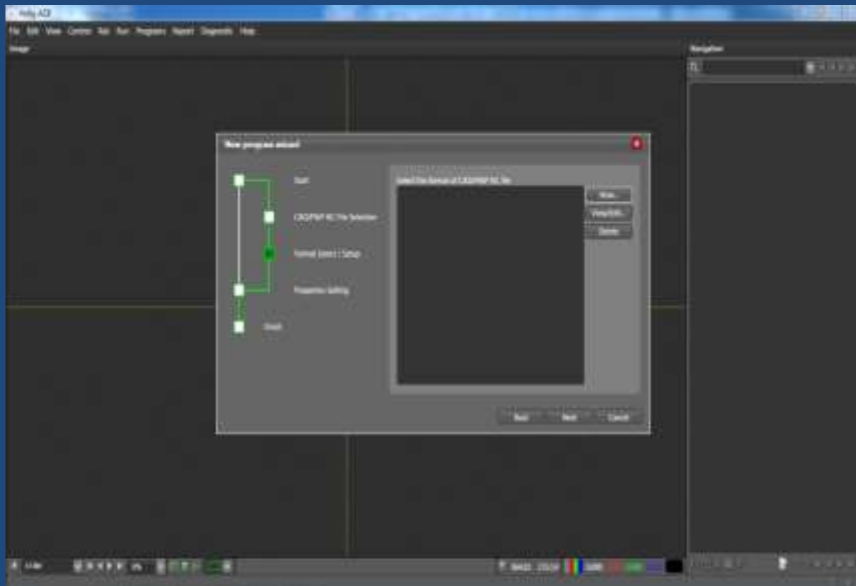
3.On the screen appear“ open file ” dialog box， find CAD file storage path， select the needed CAD file， click on the “open ” button to continue



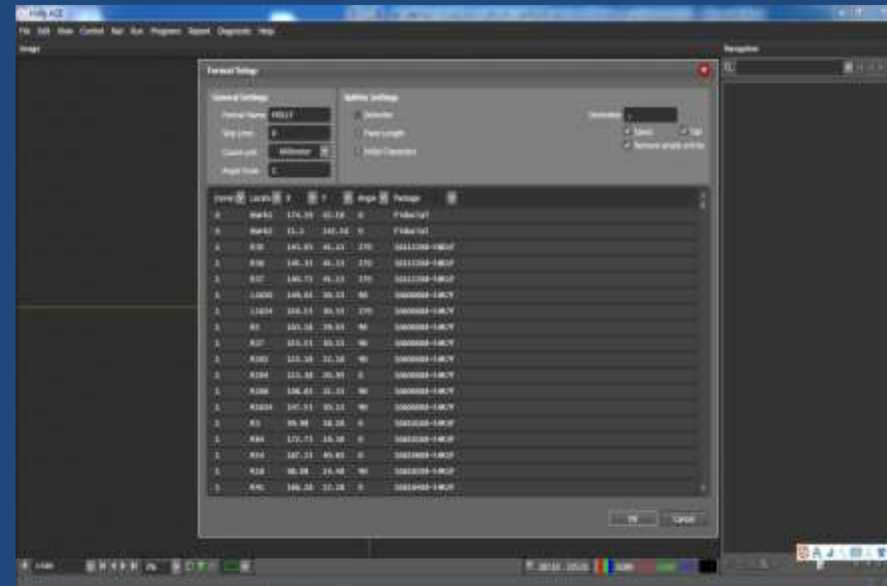
4. Click on” next “button to continue



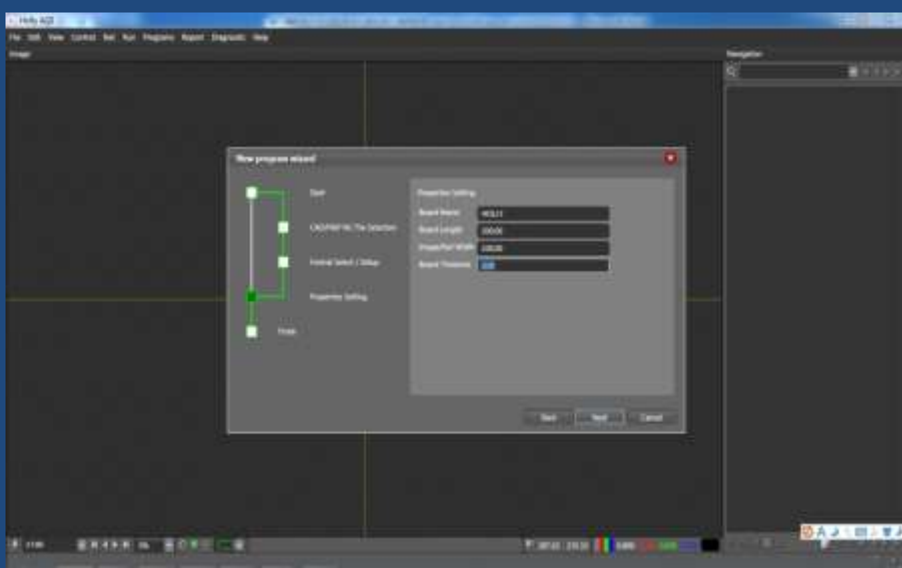
5. select CAD file format， if there is not suitable format in the list, please click on “new…”button



6. Input format name, delete unnecessary rows， select type in each column drop down list, click on “ok” to continue

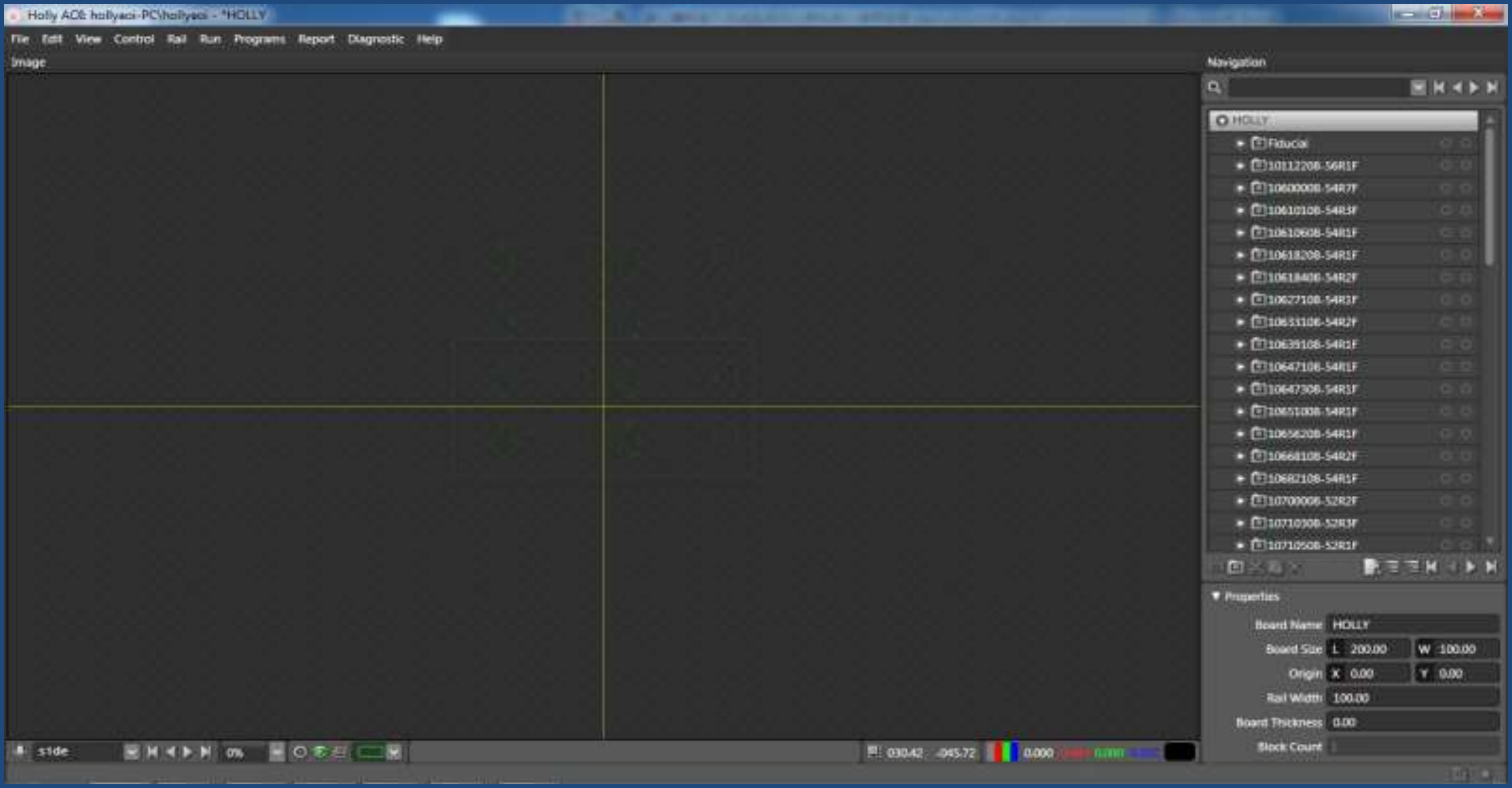


7. Input PCB board name and the length & width of PCB(the camera will capture PCB image depend on this size) , don’t changed the board thickness, its value default zero, click on “next” to finish imported CAD

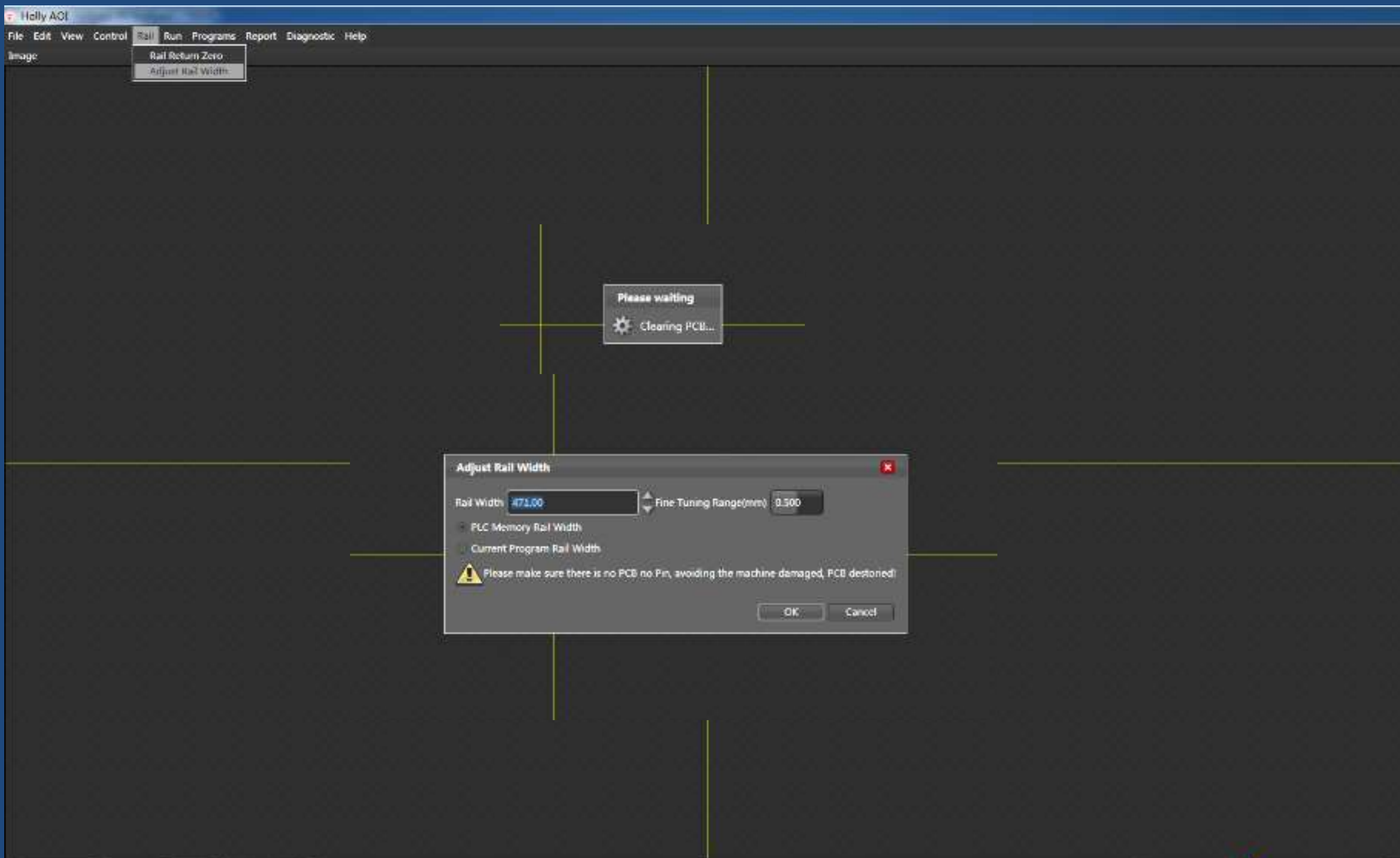


Capture image and adjust CAD layout Introduction

On the screen, it appears the following CAD general layout picture（green rectangular frame shows）and component list

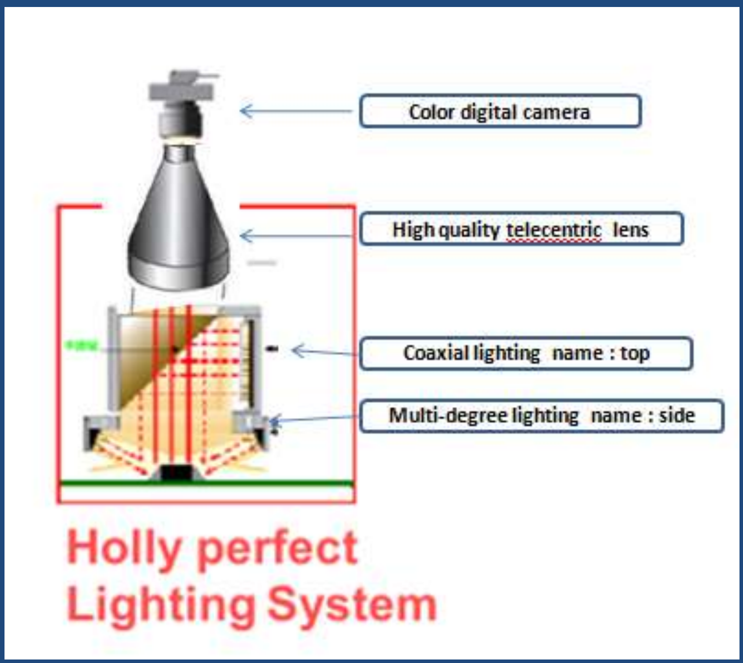


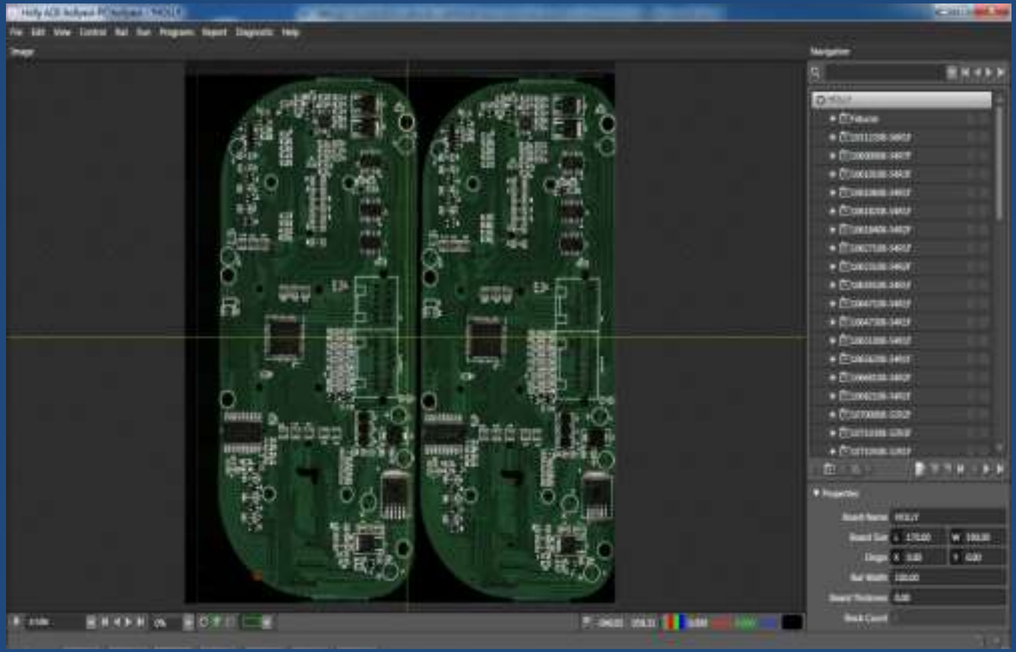
Click on “rail” select “adjust rail width”, popup “adjust rail width” dialog box , input the number of rail width refer to PCB width , then little adjust by up arrow button and down arrow button until we can place the PCB board on the rail freely



Capture entire PCB image

System alternately captures the images in way of Shuttle-Capturing with color digital camera through alternately switching on the LED lightings (top & side one by one), Due to the usage of Holly's original DISP (Dynamic multi-frame Images Seamless Patchwork) technology, we can capture several entire PCB images in one time with the highest speed.





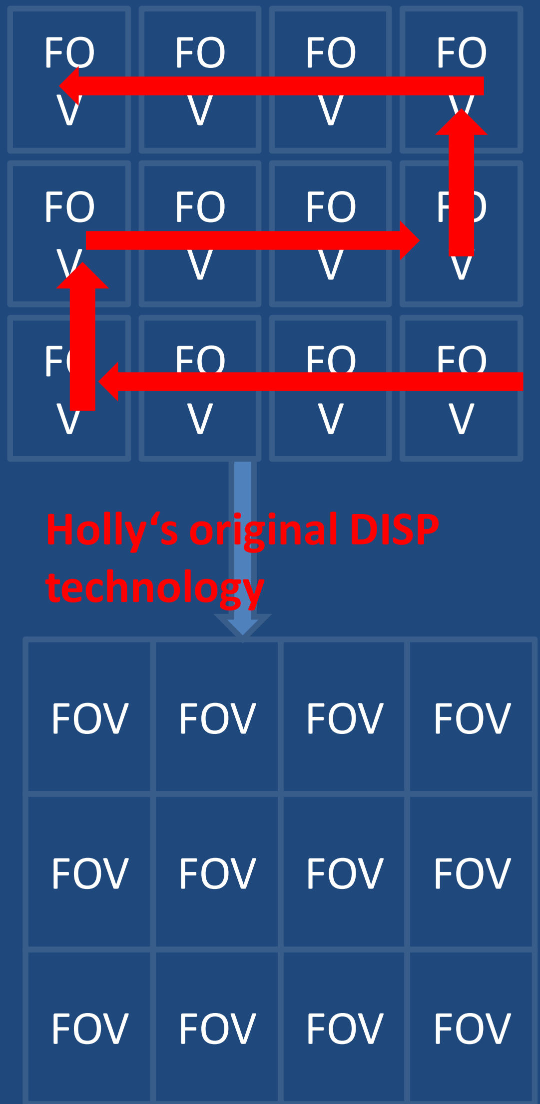
Entired PCB image

What is FOV? What is shuttle-capturing mode? What is DISP

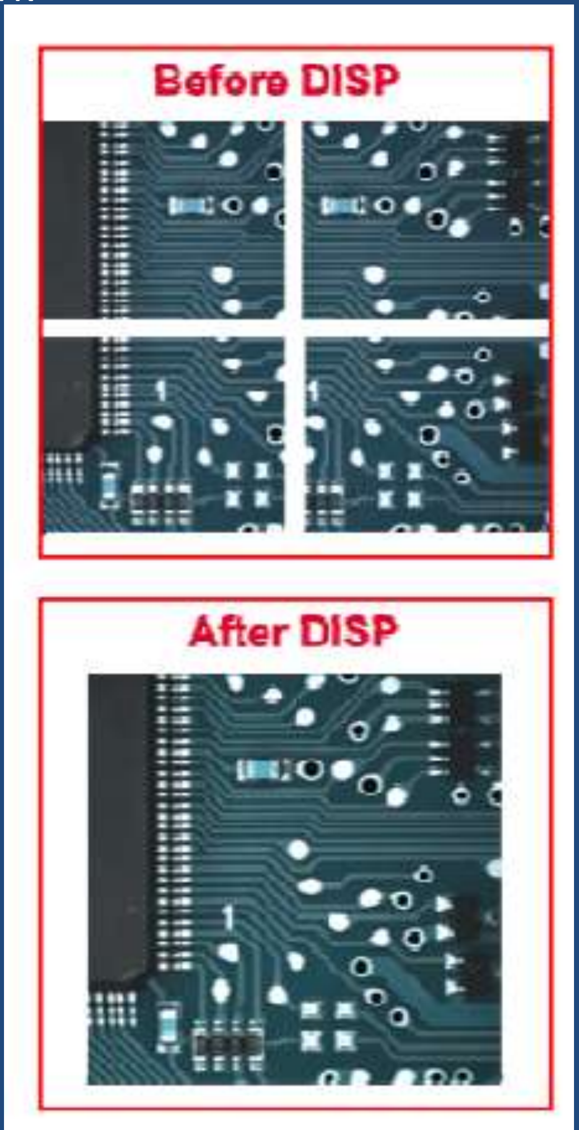
FOV “: Field of view

Shuttle-capturing： Highest speed to take several images in one time

DISP： Dynamic multi-frame Images Seamless Patchwork



The red arrows stands for the path of shuttle-capturing with camera

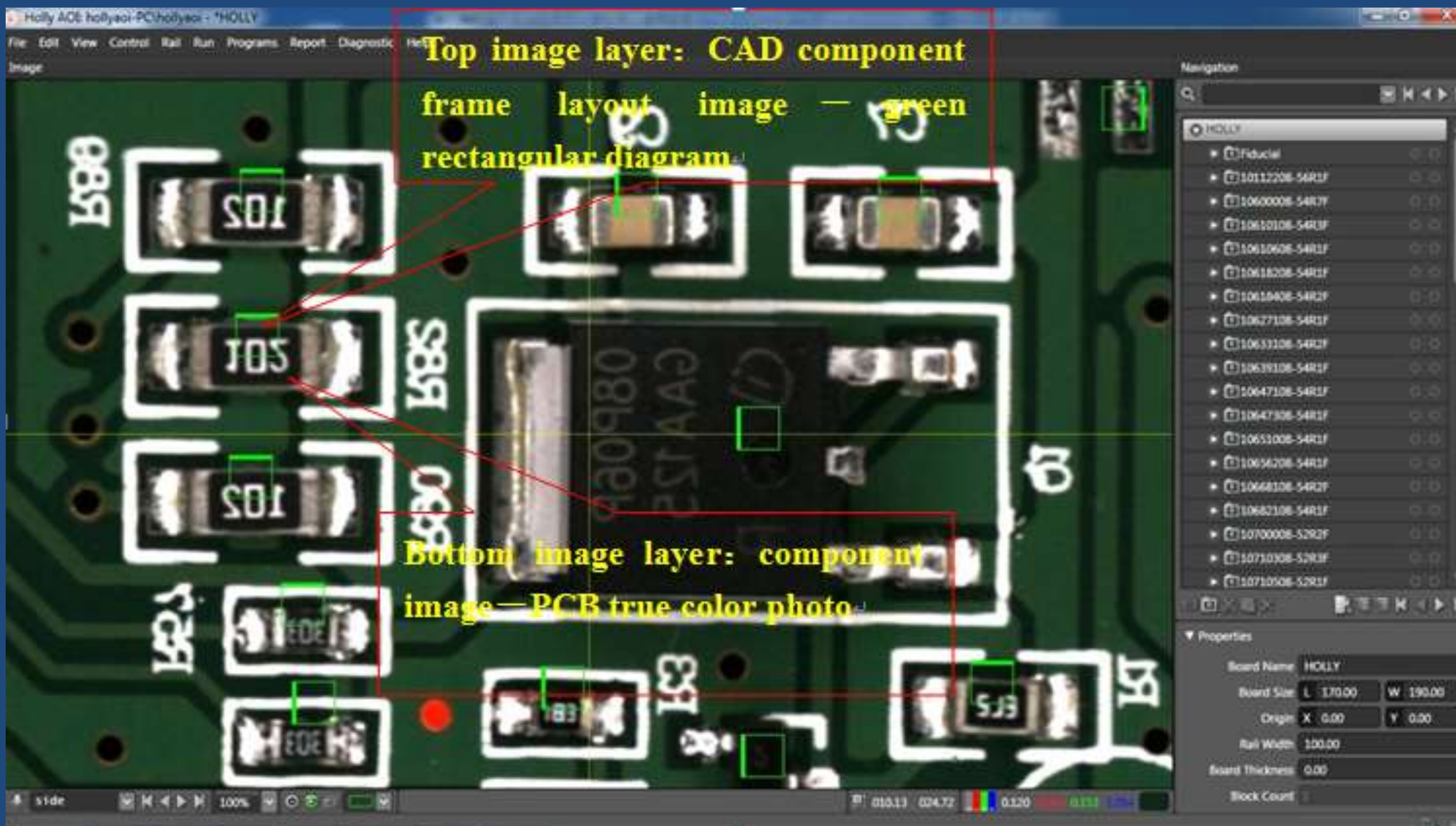


The step of adjusting CAD layout

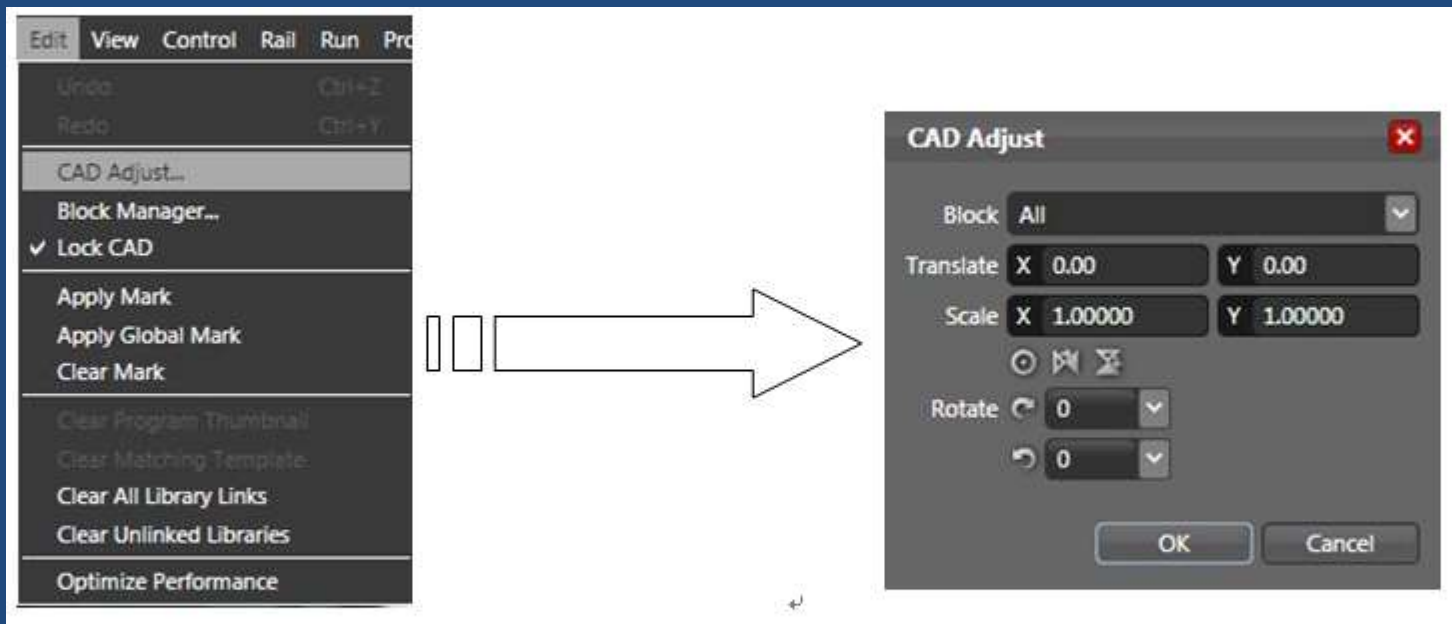
Capturing entire PCB image, then it will appear two image layer：

1). Bottom image layer：the entire board image of all components physical general layout（PCB true color photo）；

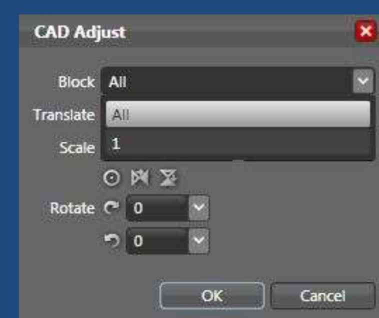
2). Top image layer：all CAD components frame layout（green rectangular diagram）



Clicking on “ Edit” menu item to select “CAD Adjust ”function item，then popping up “ CAD Adjust ” dialog box， it can adjust CAD layout by real-time response way, dispensablly trying again and again.



a. The “ Block”, select “ all “mean adjust all of cad layout frame ,select 1 stands for the first block and so on



b. Translate :this item uses for translating the X,Y coordination of all the block or selected block, modified by mouse dragging or keyboard input way.

c. This is used for zooming CAD global layout.（notice： please don’t arbitrarily change the value.）

d.  This is used for displaying CAD center of global PCB layout.

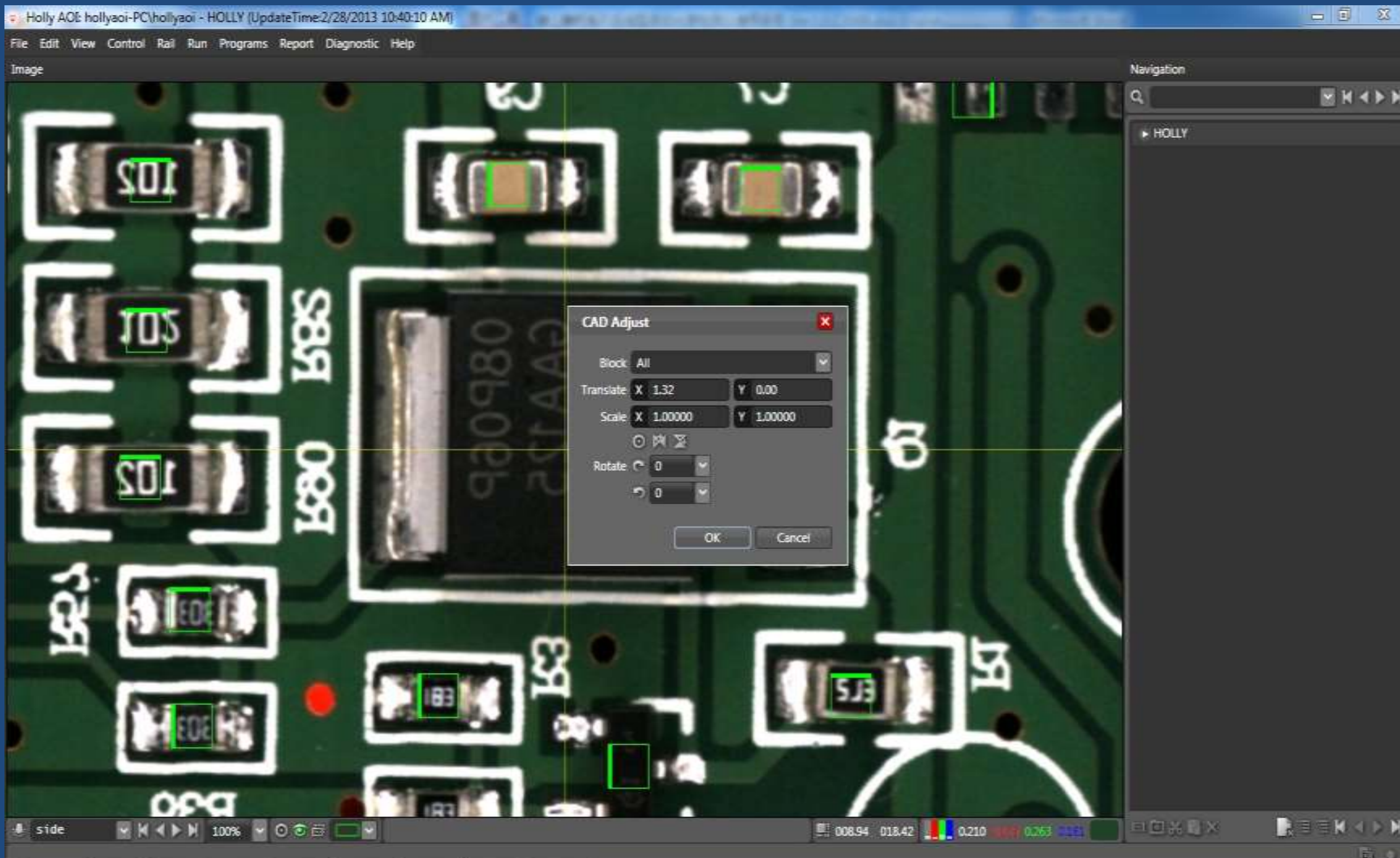
e.  This is used for fliping CAD global layout in X axis.

f.  This is used for fliping CAD global layout in Y axis.

g.  This is used for CAD layout translated by counterclockwise rotation processing. Every click is a counterclockwise rotation of 90 degrees.

h.  This is used for CAD layout translated by clockwise rotation processing. Every click is a clockwise rotation of 90 degrees.

Careful adjustment, until the two layers completely accurate superposed, click the “OK” button, complete the CAD layout adjustment.
Use the function buttons, the CAD frame layout and the real component image can be match, as follows:



The methods of making mark inspection program library

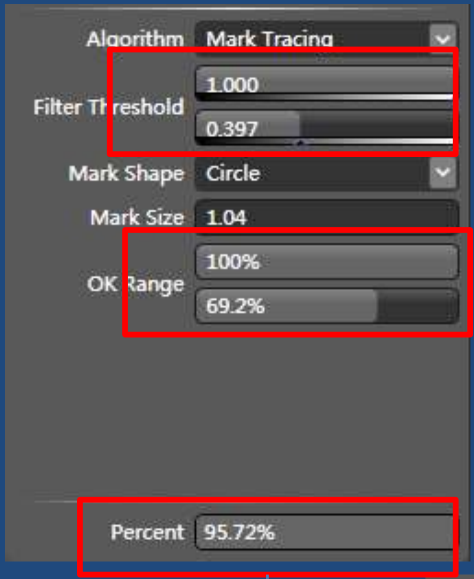
Mark tracing:

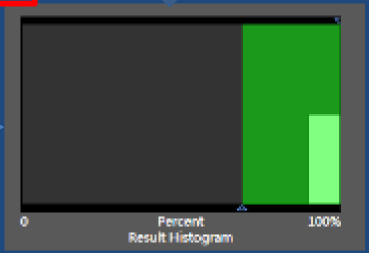
in the inspection window the designated area, through the luminance filter and Mark point shape, size is set to automatically search and locate the Mark point in a central location

Search a lighting source which has obvious luminance difference between mark and the other place which around the mark in the search box the designated area， For example， the mark luminance value about 0.4 and the other area around mark luminance value is 0.1， we need to select mark luminance value and filter the value which we don't need , the selected value ,it appears in the search Window is white effect, the filtered luminance value appears black effect, and the green circle will automatically tracing the white effect part only so that we can find the center of mark image accurately

Light source effect

Extracted effect

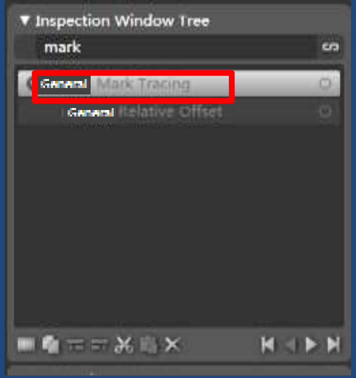




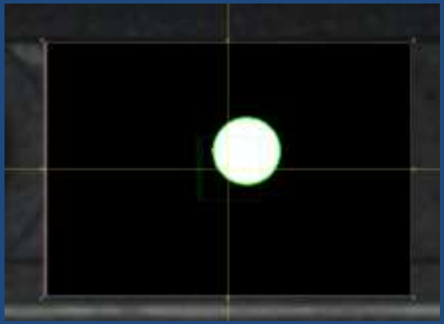
Set ok range and calculate the proportion of white effect part in the green search circle designated area

Relative offset:

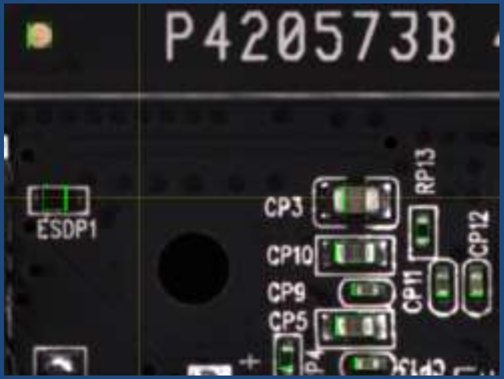
Every time we capture the image , the center of Mark coordinate and the center of mark image have relative offset, when the green circle searched mark image, it has a X,Y distance moved,
We need a algorithm to ask all component coordinates to move the same distance so that two layers completely **accurate** superposed ,and this algorithm named : **relative offset** , select it and demote it as a child window .then apply mark, you will see the mark coordinate match with the mark Image accurately , and move the mark coordinate ,apply mark again and again, you can adjust all components coordinates until superposed



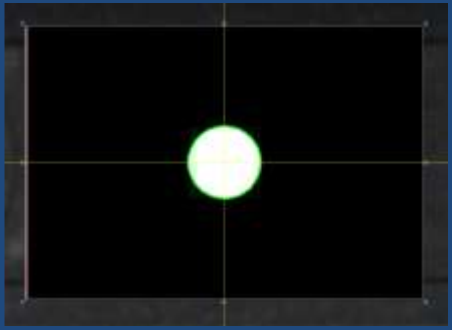
Adjust mark position to ask all components move the same distance



X,Y distance moved,



two layers completely accurate superposed



apply mark and located

The method and step of making “ Bad Mark”（bad block identification on PCB） inspection library

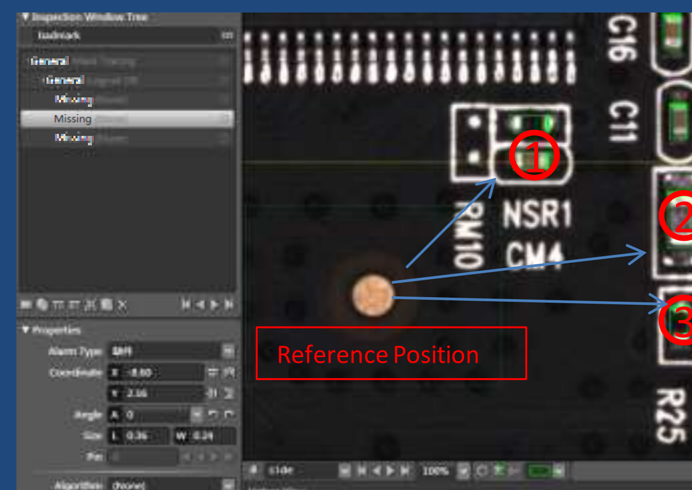
Bad Mark inspection library is mainly used for automatically shielding one bad block in all blocks on the PCB.（The bad block is caused by its poor material, also called scrap block and usually was not mounted any components on it.）
For an inspection program, the number of blocks is the same as Bad Mark number. When Bad Mark inspection window alarms on a block, the block will be considered as a bad block and the block is automatically shielded off inspecting; when Bad Mark inspection window is without alarming on a block, the block will be automatically inspected.
Notice： Bad Mark block number is the same as block number

First, searching a position which can be located by algorithm on the block, then creat badmark coordinate

The second, select algorithm “logic or” ,

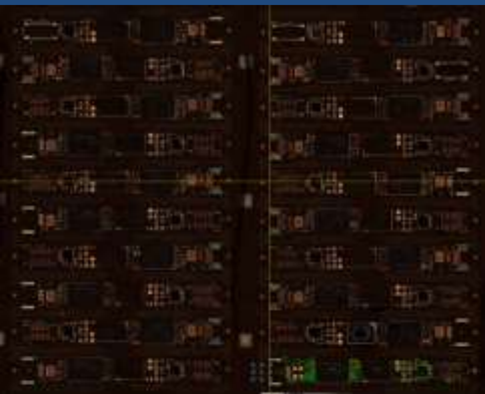
Definition: It is as parent window, as long as one or one group of its sub-windows is qualified, it is judged to be qualified

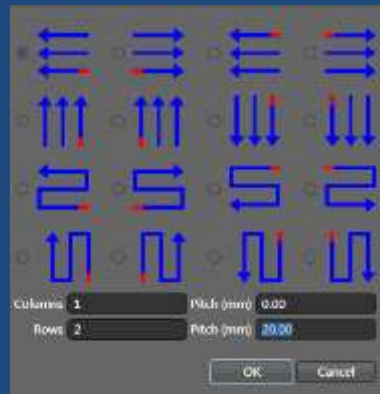
The Third , make several inspection windows to nearby component base on location which we have searched by algorithm(usually we make three windows is well), then try to make inspection algorithm to inspect “missing”, If one of sub windows is qualified , it means there maybe have components on block, it is a good block, otherwise, it is a bad block, the block is automatically shielded off inspecting

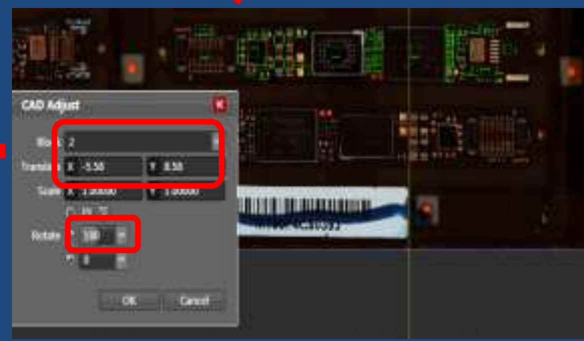


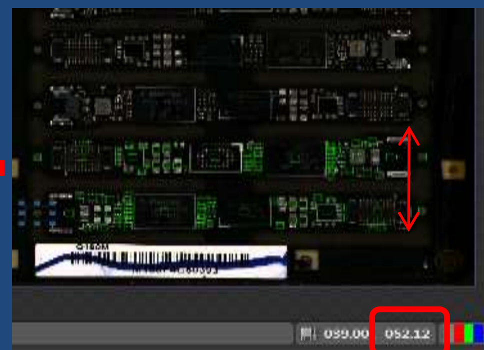
Making inspection algorithm to inspect “ missing” in three windows

Do block copy for mutil-block board copy : input the column number ,row number and distance number of them by measured , it appears multi-block coordinates which is nearly matched with components image ， apply mark, you can see the other coordinates and component images accurately superposed



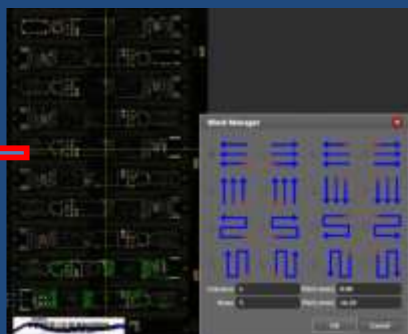






Measure distance between Y direction

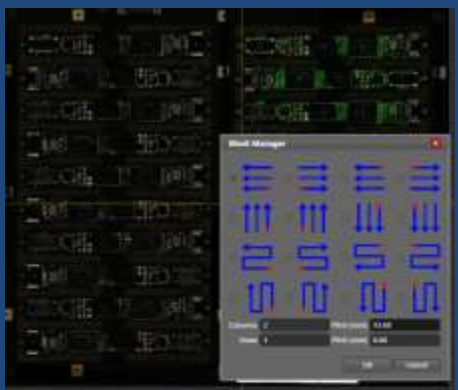
Measure distance between Y direction



Input Rows numbers data and distance

Apply mark

Measure distance between X direction

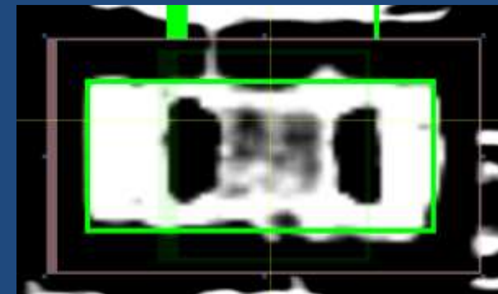


Input columns numbers data and distance

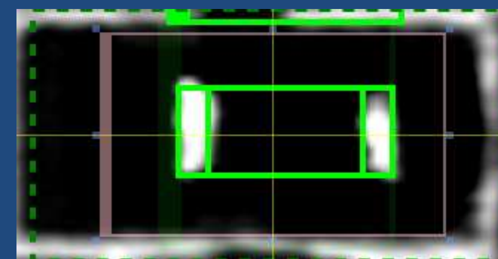
Apply mark

FINISH BLOCK COPY

Chip tracing:
in the inspection window designated area, according to the specified component length and width size, the size of electrode, to automatically trace the positioning component position

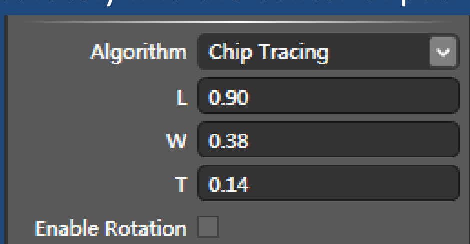


Located the center of pad



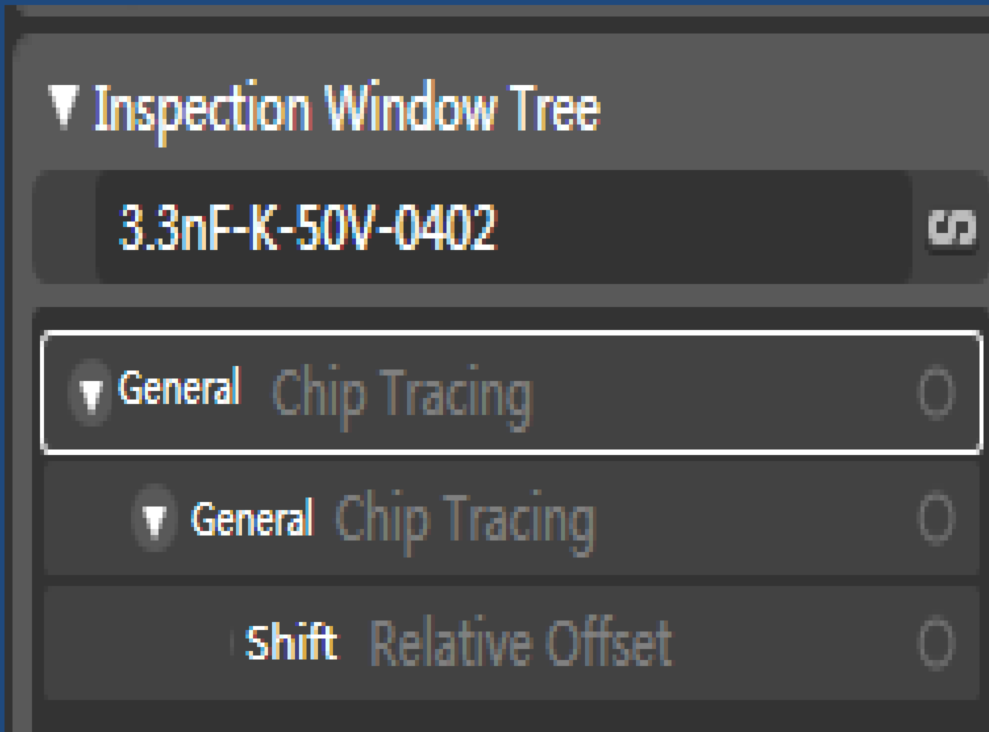
Located the center of component

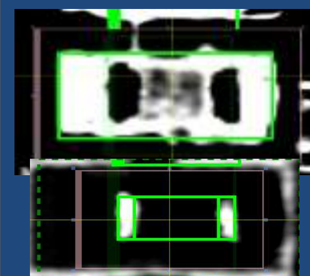
Sometimes when we capture image ,the center of coordinates is not the same center of the component, so we need an Algorithm to research the center of the pad. it need a light source which the shape of pad appears white light effect and the other place in the search window appears black effect .then we input the lenth and width here . the green rectangular frame will automatically search the center of the pad so that we can accurately find the center of pad

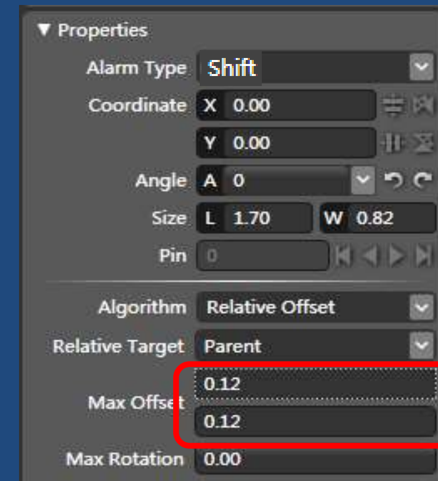


Relative offset:

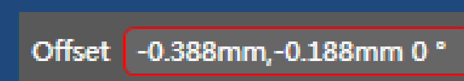
Definition: can be set relative to the parent window, components or PCB offset out put, we need to located the center of pad and component by the way of location methods





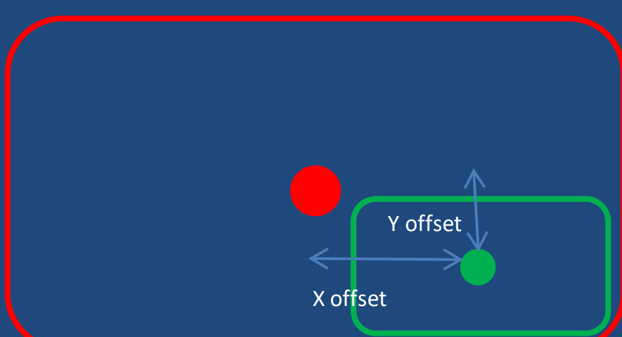


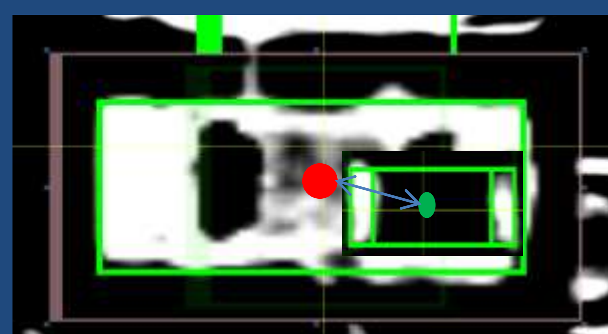
Set max offset of X ,Y direction,
If beyond range it will alarm shift



X offset :0.338>0.12

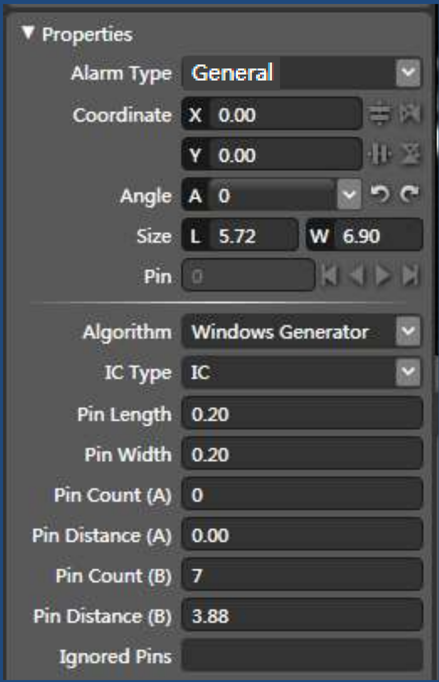
Y offset :0.188>0.12

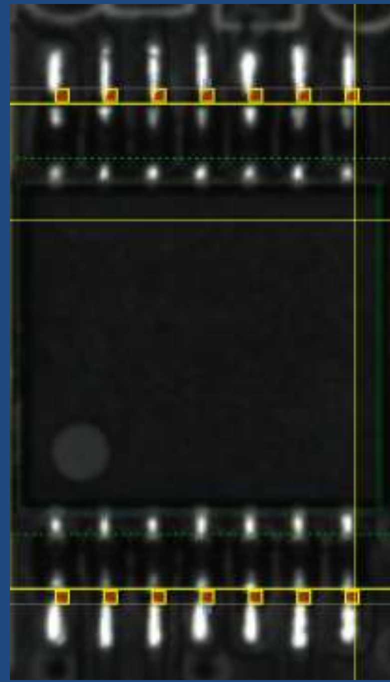


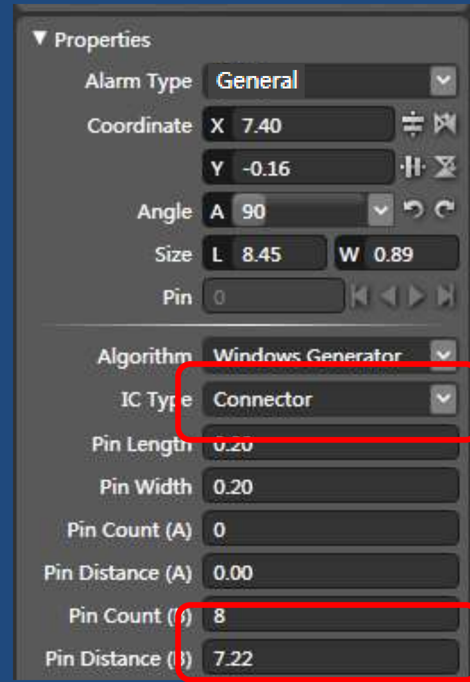


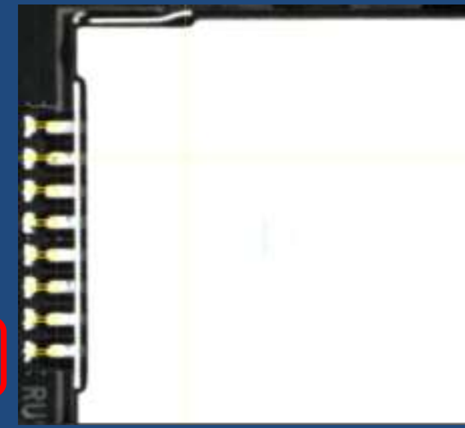
Windows generator:

Definition: To produce a series of inspection windows automatically in the inspection window area designated, with real-time response mode,







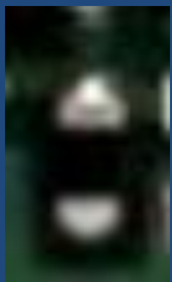


Select IC type of " IC", it means Seven pairs of windows

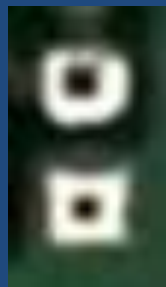
Select IC type of "connector", Eight windows for single side

Input pin count （A or B) and pin distance(A or B) which we need , dragging the windows, make the red point into the inspection area, add new , demote it as a child window, the inspection windows automatically move to the designated area, in this area , we can inspect no solder ,copper leak, or dry joint

HSV Extraction:
in the inspection window designated area, obtain consistent with HSV color qualified pixel number accounted for a percentage of the total data window pixel



Good



Missing

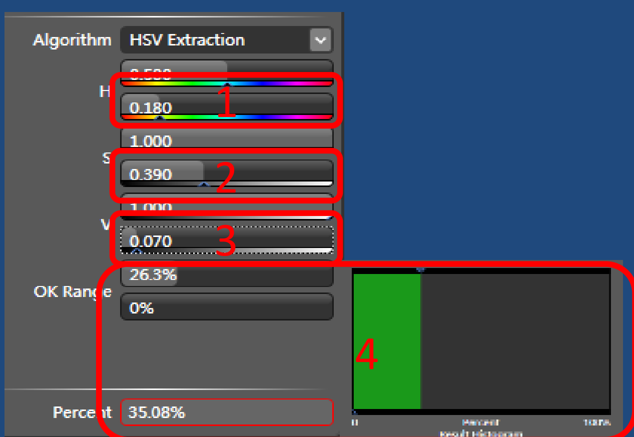
Different？ can we consider that when we see green color it means missing. we can make algorithm named color extraction(HSV)

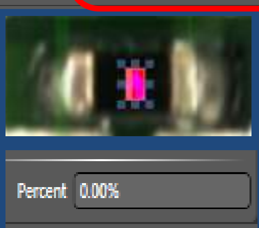
In fact image

Step 1： Select color (H lower) what we need, it appears white effect in the inspection window, but sometimes the object also appears white effect

Step 2: Select color (S lower) and make it fuller

Step 3: Select color (V lower) if we still have not enough effect to distinguish between green and othe area， fill it again until we can see a clear shape , and dragging the window to small





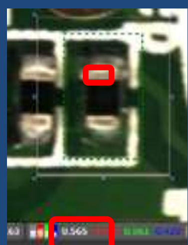
Step 4: when we get extracted effect we need ,we must setting OK range to **distinguish good or not good**. **Percentage** here automatically computed the **proportion of green color** in the inspection window . certainly ,we hope **the proportion** of green color the less the better right ? so we set ok range by left area 。

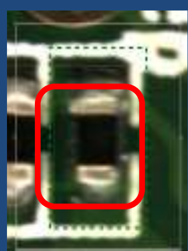
Luminance Extraction:

Definition： In the inspection window designated area, obtain with the specified number of pixels of the qualified luminance accounted for a percentage of the total data window pixel,

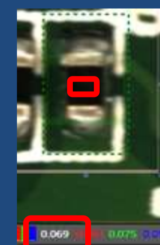
For example，the way of inspecting defect of missing by algorithm of Luminance Extraction

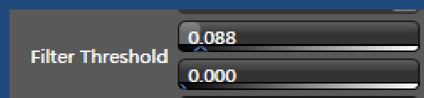
it means the luminance value in the inspection window which we selected ,it appear bright effect, dragging the window like this , we know that the electrode luminance value is greater than other place , we filter luminance value until we can distinguish between electrode area and black area, ，if we select left area , it means we hope that the proportion of black area the more the better in the inspection window, so we setting ok range by right area， if shift， the proportion of black area will be 霸占seized by electrode, when it 超出**exceed** our setting range,it will be alarm shift

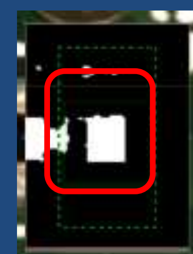


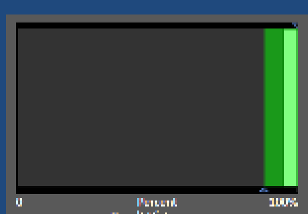


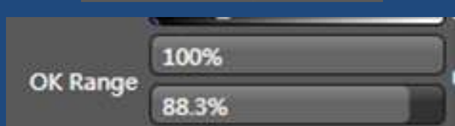
Select black luminance value , it appear white effect



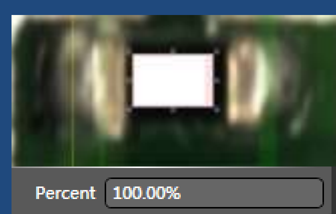




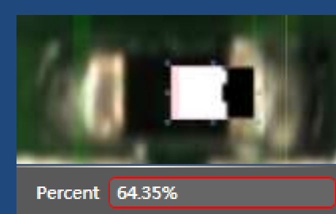




Set OK Range



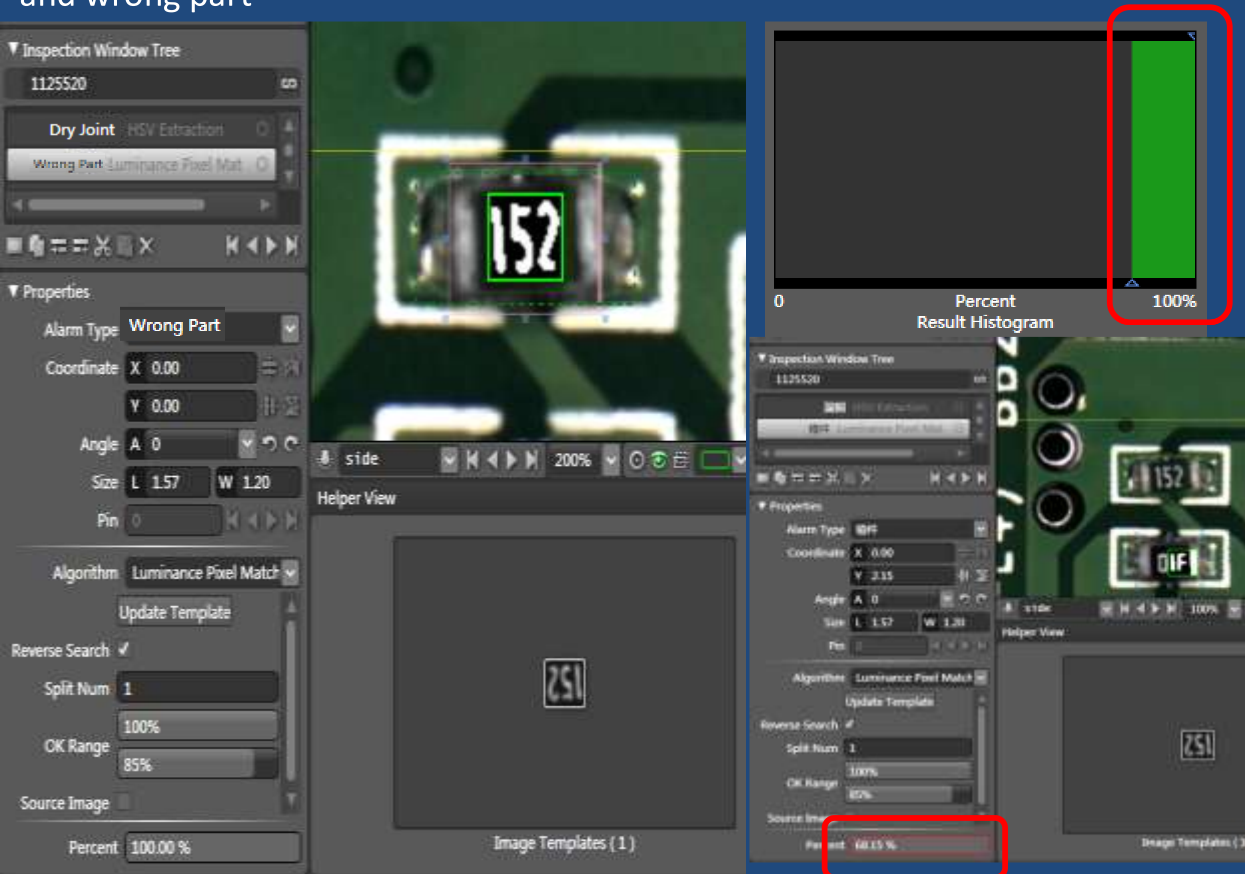
good



If Shift， alarm

Luminance Pixel Matching(OCV)

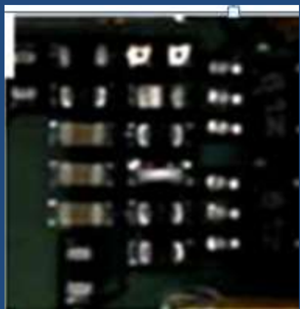
Definition: in the inspection window designated area, automatically search target(characters or graphics) most similar with sample image, and calculate the percentage of similarity degree , it is used for inspecting the defect of polarity and wrong part



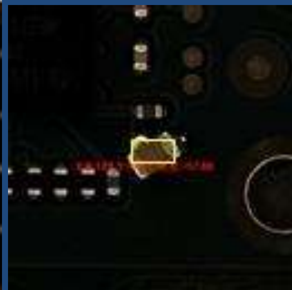
For example, if the sign 152 change to 01F,it calculate the similar degree depend on 152 (Image Template) , the percentage result is 68.15% less than 85%, alarmed wrong part

Steps: click on “update Template” , update a printing sign as template image (if component without inspecting defect of polarity, you must not click “reverse search”), and then set ok range by right area ,it will automatically calculate the percentage of similarity degree , if beyond range we set before, alarmed

Luminance projection Min span

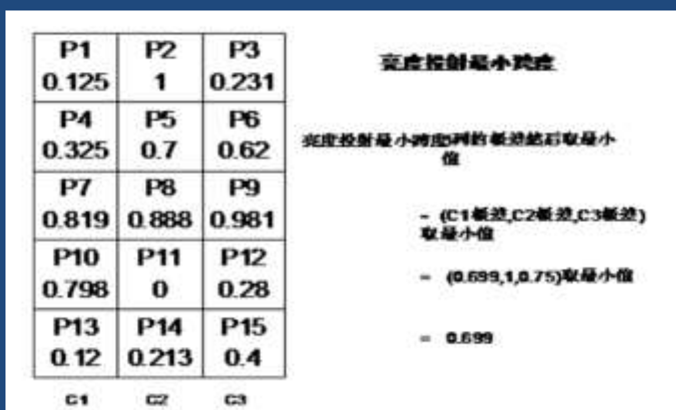


Billboard/ sidelift



shift

bridge



P1 0.125	P2 1	P3 0.231
P4 0.325	P5 0.7	P6 0.62
P7 0.819	P8 0.888	P9 0.981
P10 0.798	P11 0	P12 0.28
P13 0.12	P14 0.213	P15 0.4
C1	C2	C3

亮度投射最小跨度

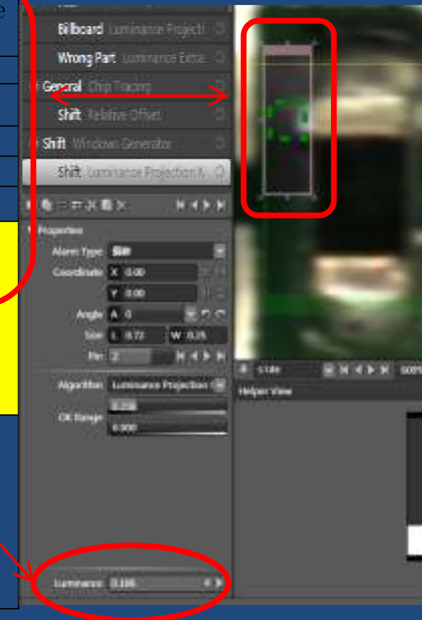
亮度投射最小跨度即列的极差然后取最小值

= (C1极差,C2极差,C3极差)取最小值

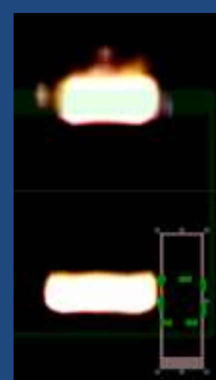
= (0.699,1,0.75)取最小值

= 0.699

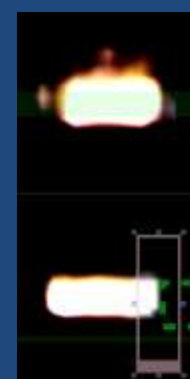
pixel luminance value	line 1	line 2	line 3	line 4
	0.02	0.26	0.03	0.3
	0.123	0.26	0.01	0.3
	0.9	0.26	0.00	0.3
	0.034	0.2	0.01	0.3
	0.5	0.275	0.08	0.6
line span value : (Calculate the defference between the MAX value and Min value in the length line direction)	0.88	0.075	0.08	0.3
the computed result of luminance value is the minimum line span value which is obtain from all the line span value	0.075			



Similar compare effect image

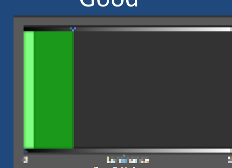


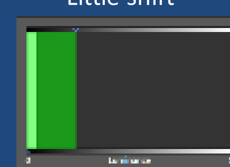
Good



Little shift

Shift alarm



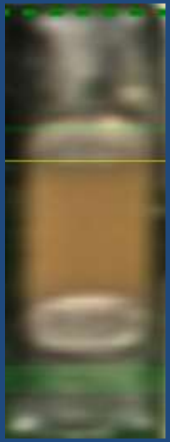


Luminance Mean

Definition： in the inspection window designated area, calculate all the pixel brightness to get average

For example， the way of inspecting defect of wrongpart by algorithm of Luminance Mean

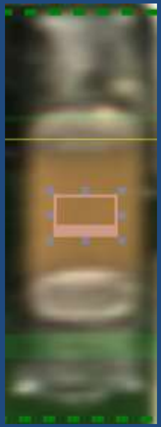
Luminance Mean				
pixel luminance value	0.3	0.1	0.3	0
	0.2	0.2	0.1	0
	0.4	0.5	0	0
	0.9	0.1	1	0.3
	0.3	0.6	0.6	0.6
add all number and caculate	6.5			
Get the average luminance mean value	6.5/20=0.325			

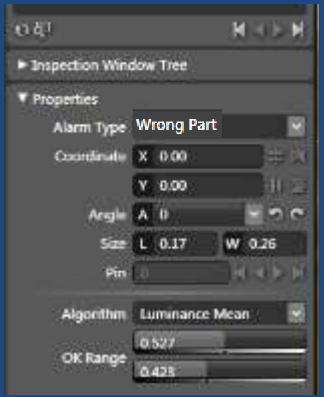


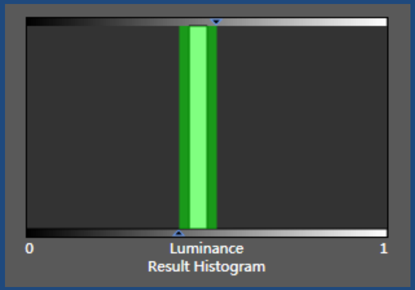
If wrongpart， it may be a resistor

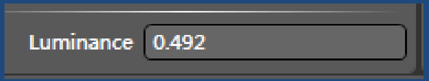
What is different or feature between them???

all Luminance > all luminance

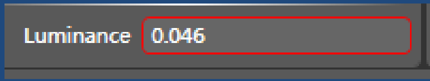








>



Dragging the inspection window to the surface of capacitor , click on “inspect” icon ,it will directly automatically calculate the average value of resistor , you just bound the value is ok

0.423<0.492<0.527

In the range , good

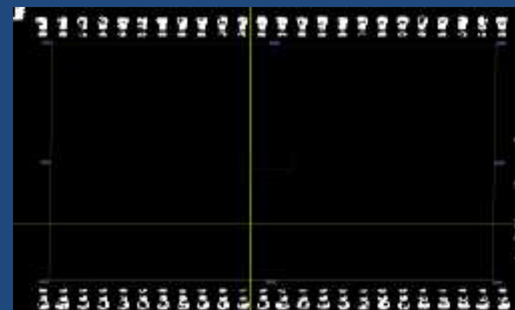
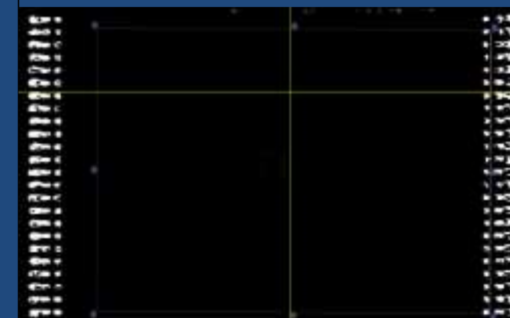
0.046<0.423

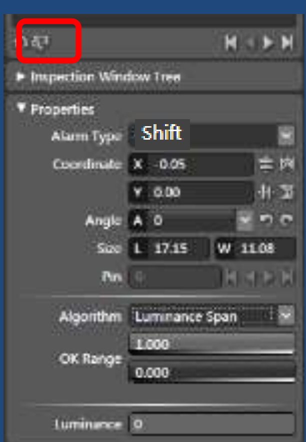
Less than the range of minimum value, NG

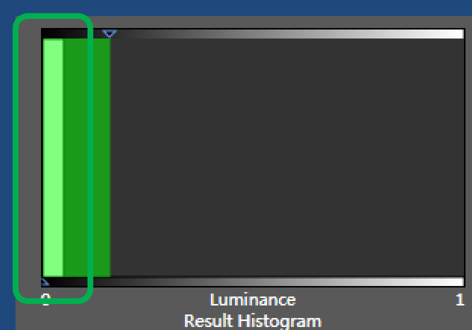
Luminance Span

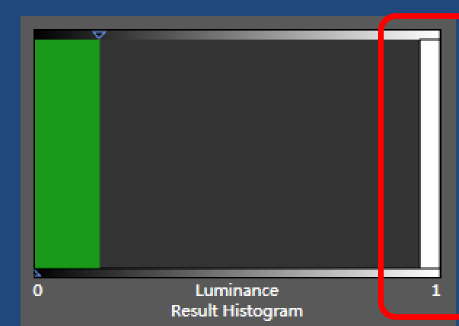
Definition： the inspection window designated area, calculate all the brightness of the pixel to get the difference between maximum and minimum values

For example， the way of inspecting defect of shift by algorithm of Luminance Span

Luminance Span						
	0.3	0.1	0.3	0		
	0.2	0.2	0.1	0		
	0.4	0.5	0	0		
	0.9	0.4	0.1	0.3		
	0.3	0.6	0.6	0.6		
calculate all the brightness of the pixel to get the difference between maximum and minimum values	maximum value:0.9				0	1
	minimun value:0				0	0
	Difference equal to 0.9-0=0.9				Luminance 0	Luminance 1





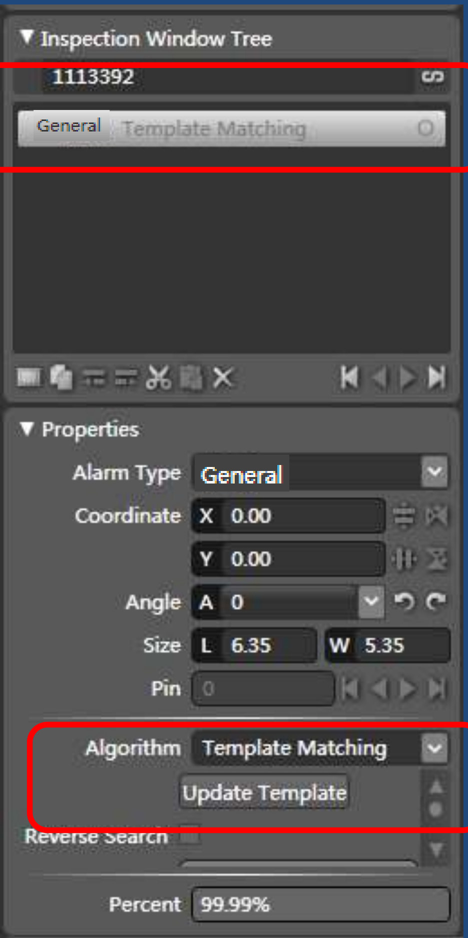


Beyond the range
Alarm “shift”

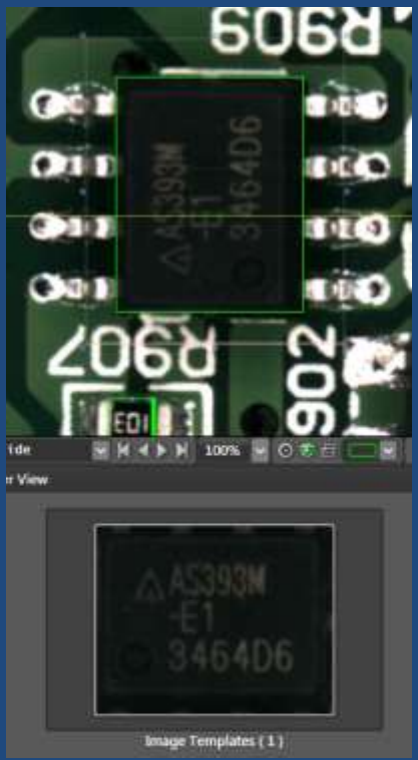
Searching a lighting source which the luminance value is as the same value as possible, in the inspection window ， click on “inspect” icon ,it will directly automatically calculate the difference between maximum and minimum values

Template Matching（RGB）

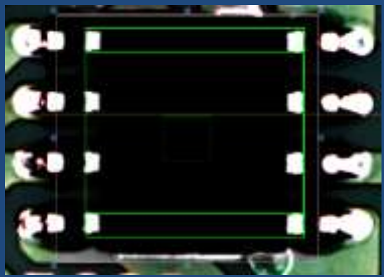
Definition: in the inspection window designated area, automatically search target(graphics) most similar with sample image(color pixel), so that we can find the center of object and make inspection algorithm depend on its coordinates, sometimes we can't find the light source which appears white effect, we can use this algorithm



Click on update template , and update image which we need to used located



Located by sample image

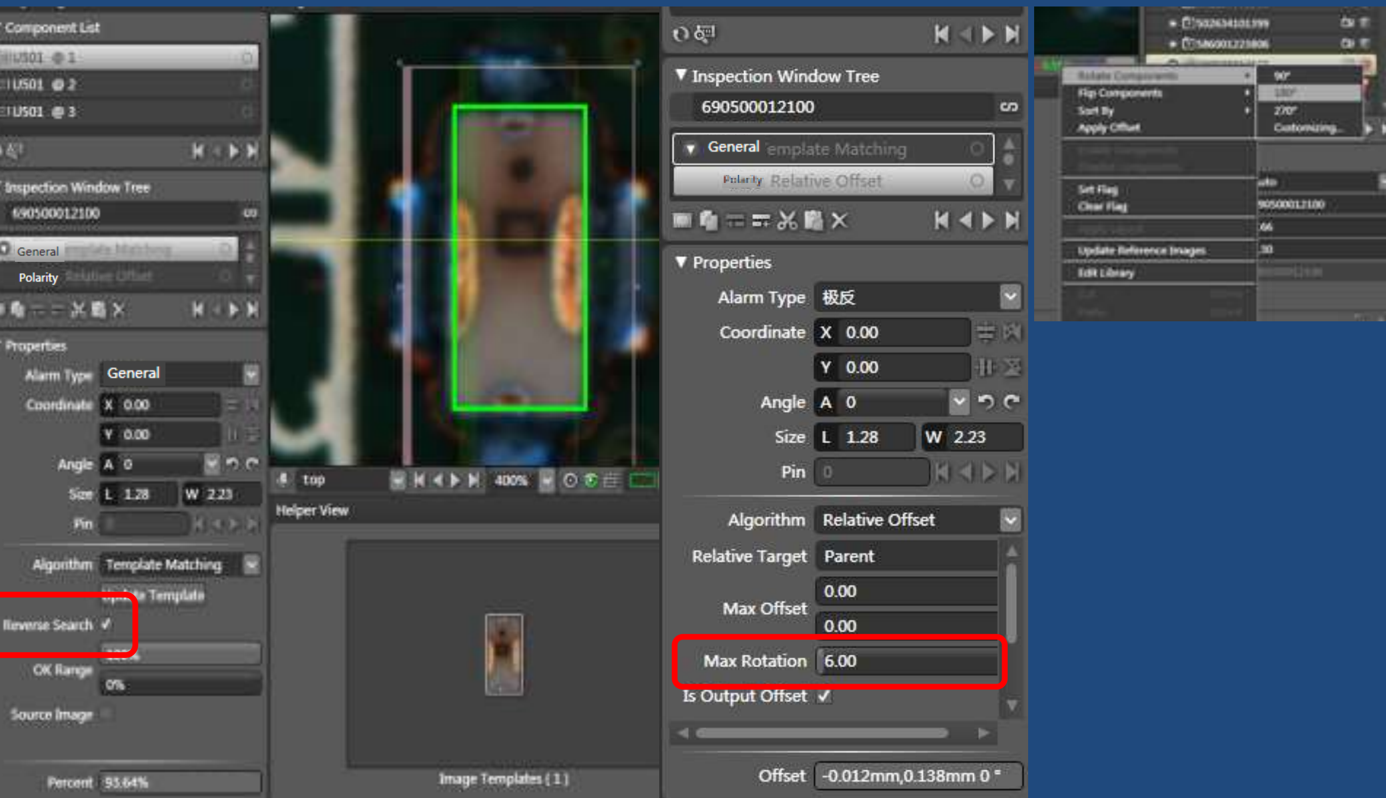


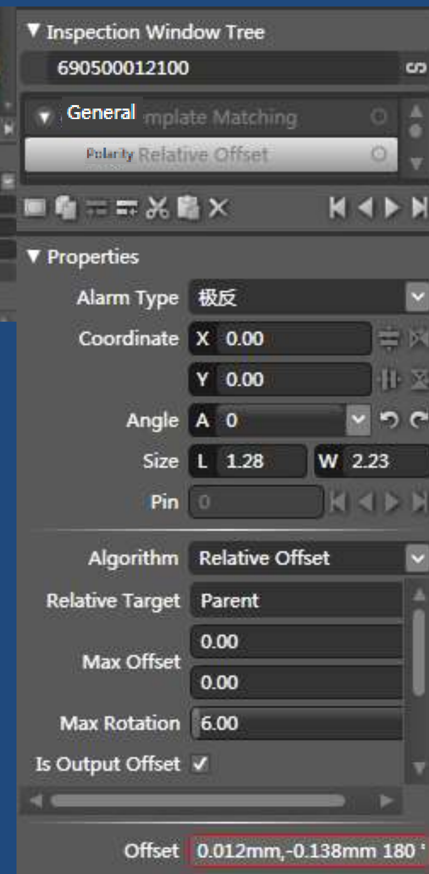
Another location method by light source

Chip tracing

Template Matching（RGB）

It has another usage for inspecting the defect of polarity

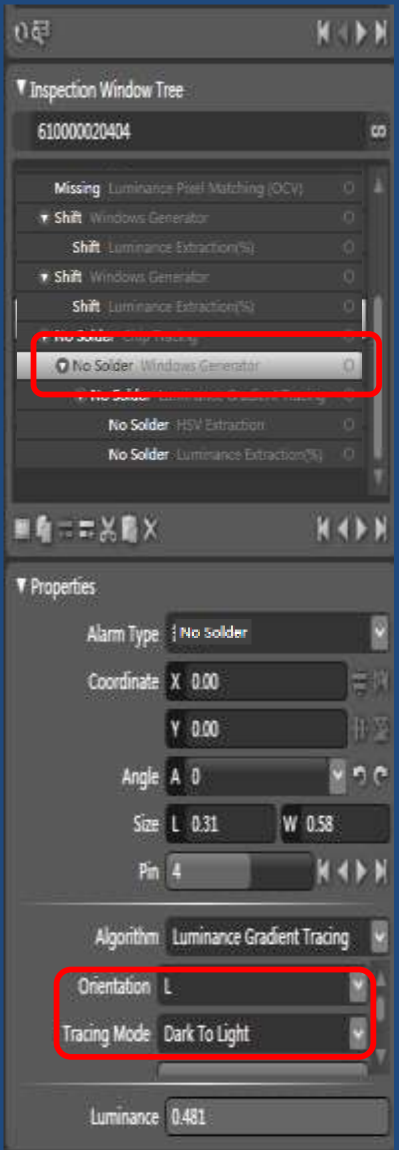




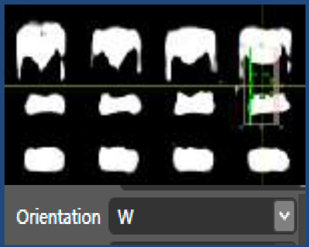
Sometimes， the component have not obvious feature position for inspecting polarity, we can use this method,
Just select “reverse search” & input the MAX rotation degree range (less than 180 degree) , it means , in the inspection window designated area, automatically search target(graphics) most similar with sample image, if polarity, it will automatically caculate the image which rotate 180 degree ,beyond our range , alarmed polarity

Luminance Gradient Tracing

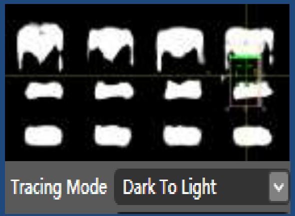
Definition: To automatically trace the biggest change of brightness of the single boundary in the inspection window designated area by two direction choice, (long or short direction), and by two tracking mode (light to dark or dark to light) , for example ,to inspect the defect of IC no solder

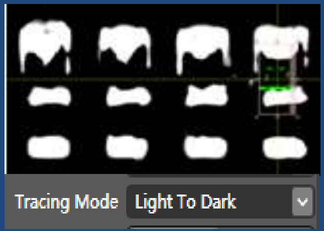


Effect for two direction choice



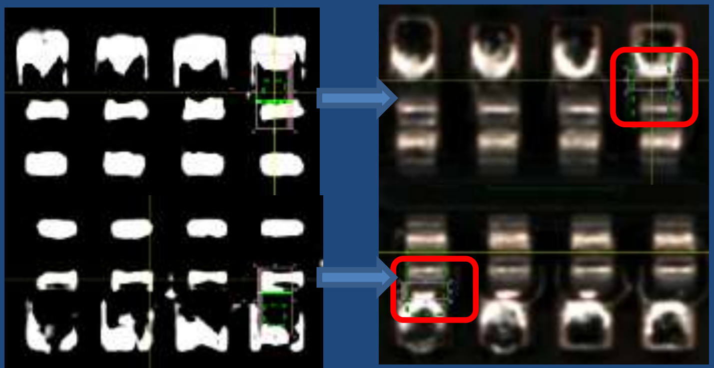
Effect for two tracking mode





Search a lighting source as top of image , select Direction choice and tracking mode which you need

Different??? Window Shift

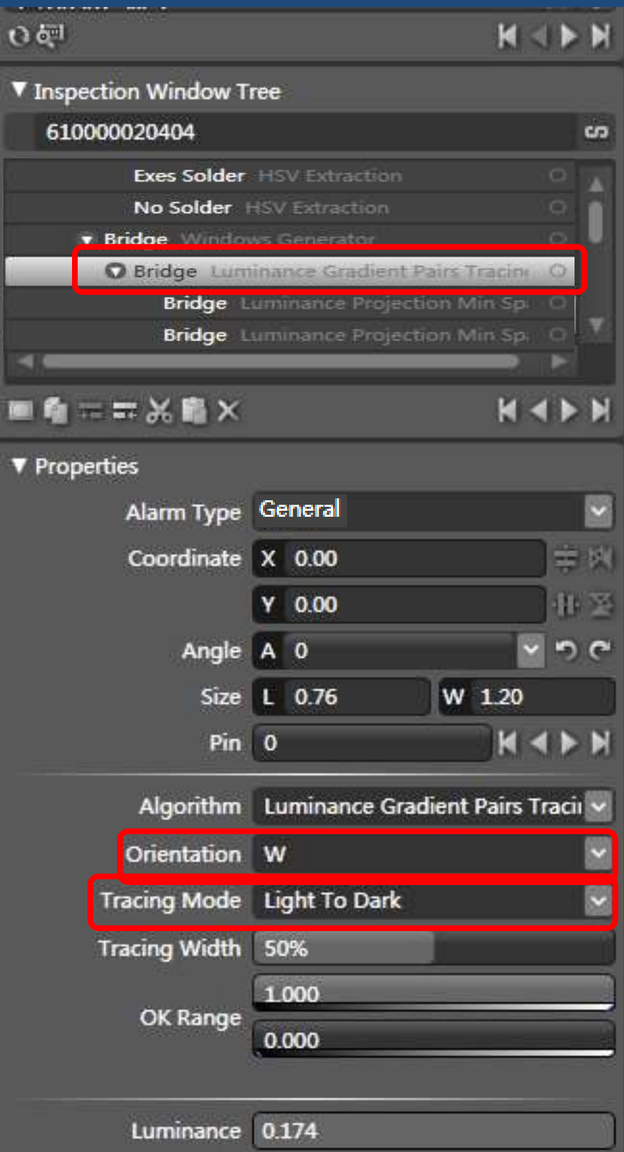


Back to the boudray

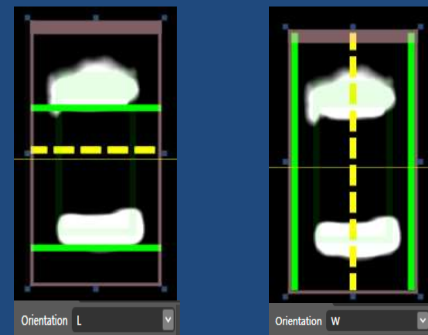
Sometimes the inspection window is not the center position which depends on the component location , we use this algorithm, it will automatically search the boundary nearby the end of pin so that we can accurate make inspection algorithm to inspect defect of no solder , dryjoint

Luminance Gradient Pairs Tracing

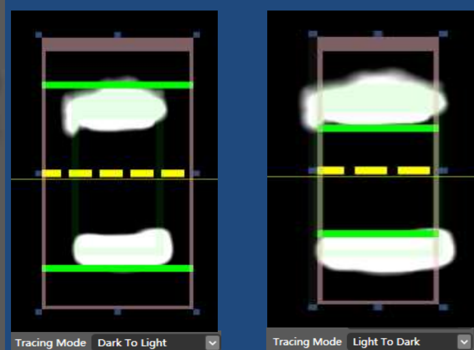
Definition: To automatically trace the biggest change of brightness of the pair boundary in the inspection window designated area by two direction choice, (long or short direction), and by two tracking mode (light to dark or dark to light), used to windows automatic tracking and positioning, for example ,to inspect the defect of IC bridge



Effect for two direction choice

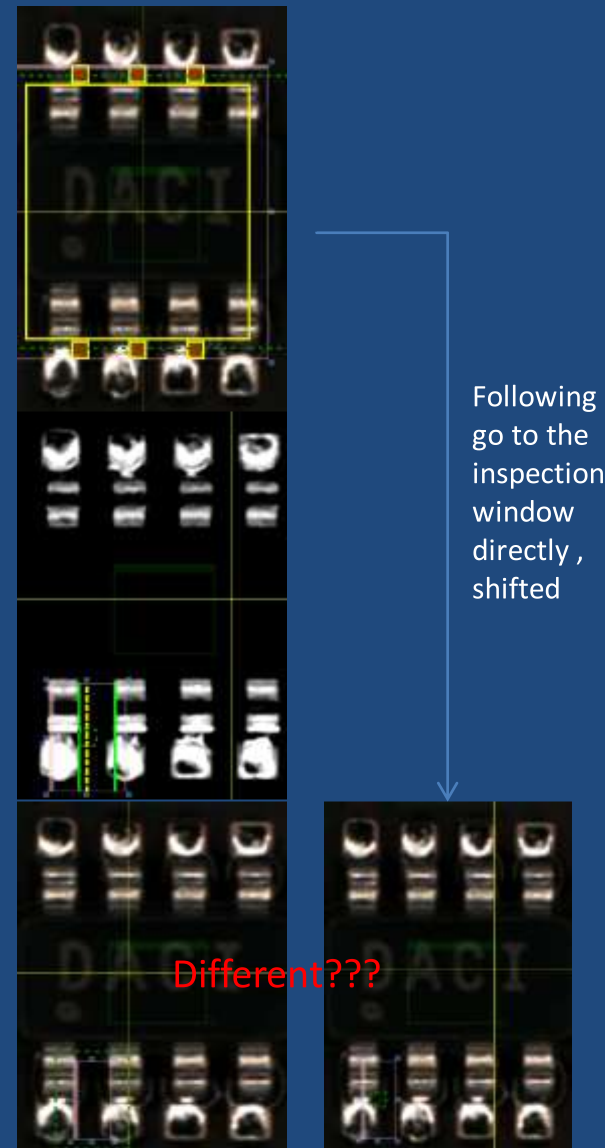


Effect for two tracking mode



Search a light source as top of image , select Direction choice and tracking mode which you need

Sometimes the inspection window is not the center position which depends on the component location, if we use this algorithm, it will automatically search the center so that we can accurate make inspection algorithm with designated area

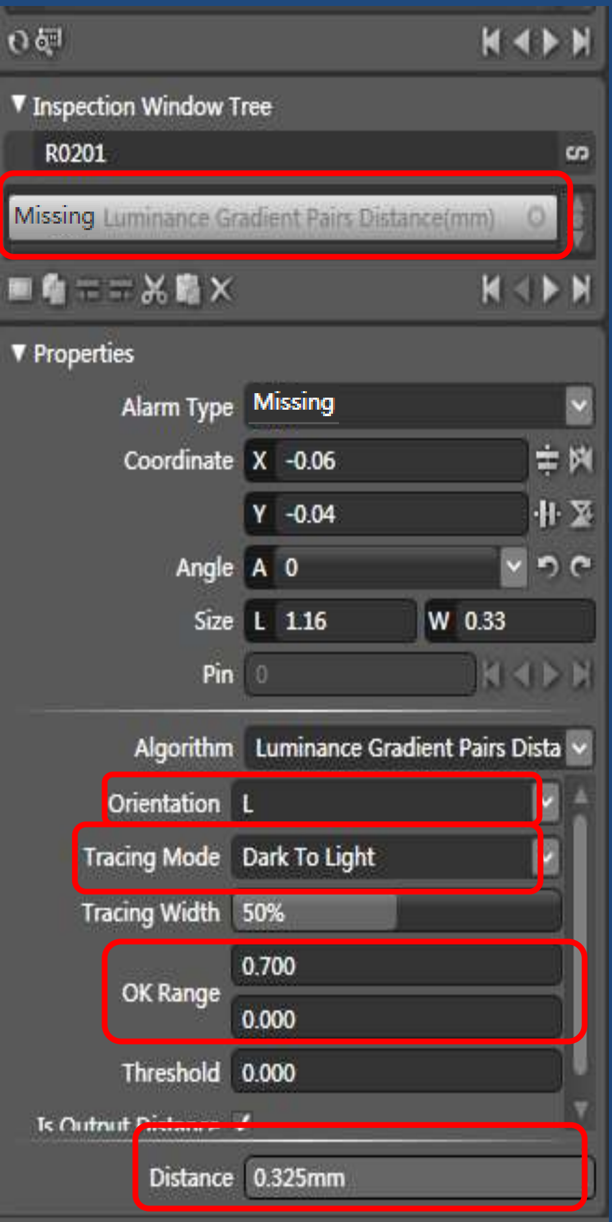


Back to the center

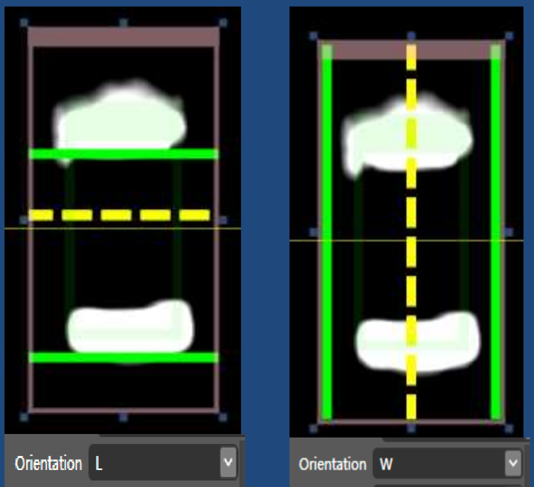
Window Shift

Luminance Gradient Pairs Distance (mm)

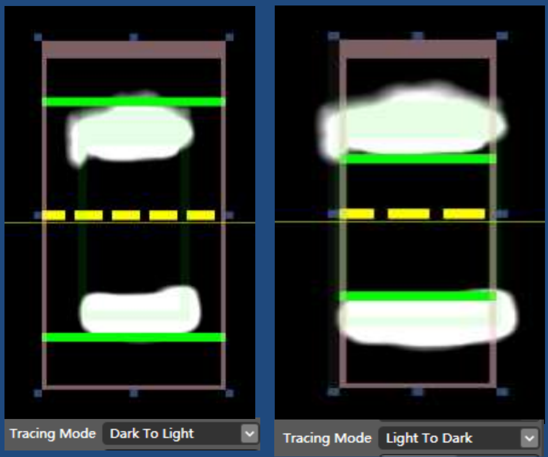
Definition: To automatically trace the biggest change of brightness of the pair boundary in the inspection window designated area by two direction choice, (long or short direction), and by two tracking mode (light to dark or dark to light), calculate the distance of two boundary compared to the distance of the inspection window



Effect for two direction choice

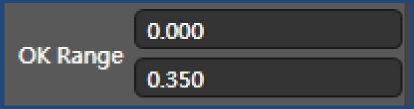


Effect for two tracking mode



Difference

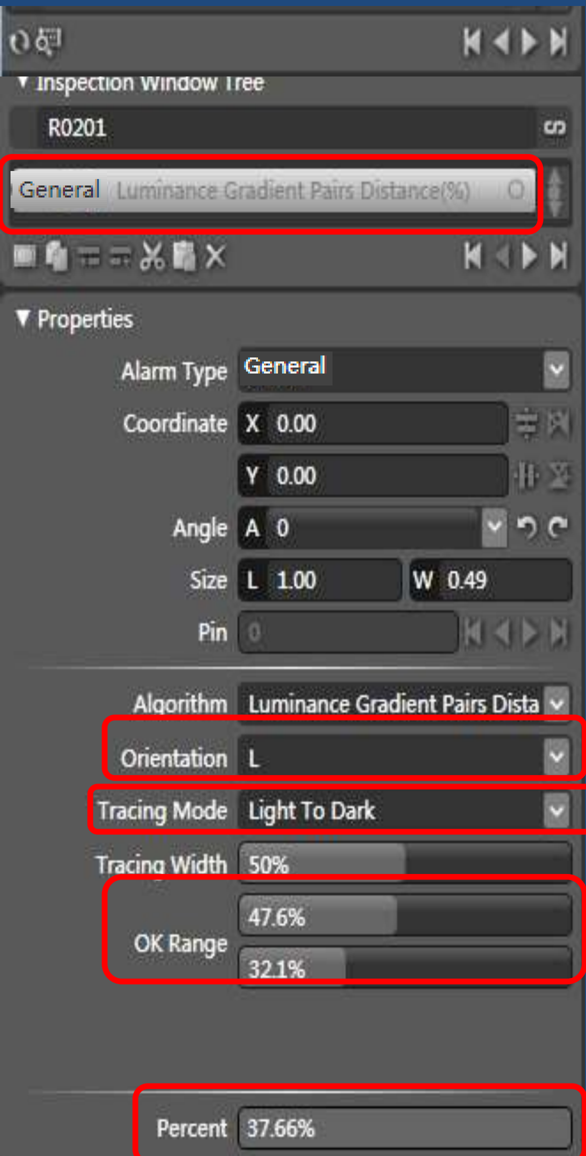
Search a light source as top of image , select Direction choice and tracking mode which you need ,set ok range



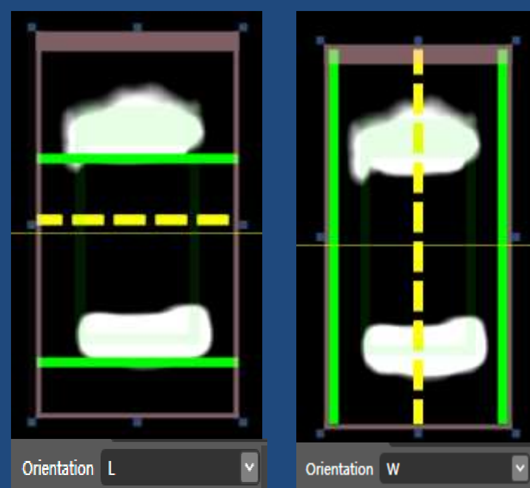
If wrong part , the distance between two green line is 0.200 less than 0.35, it will alarm

Luminance Gradient Pairs Distance (%)

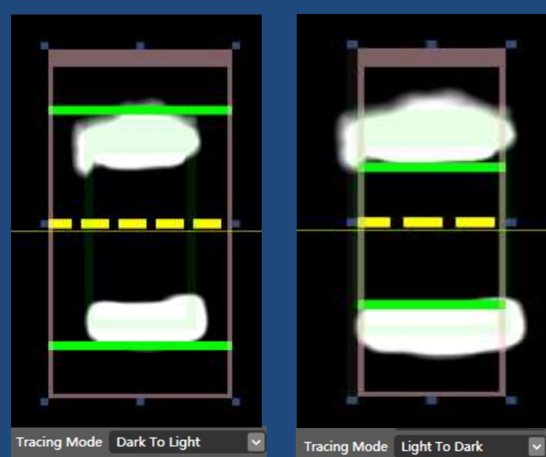
Definition: To automatically trace the biggest change of brightness of the pair boundary in the inspection window designated area by two direction choice, (long or short direction), and by two tracking mode (light to dark or dark to light), calculate percentage of distance between two boundary compared to the line length of the inspection window

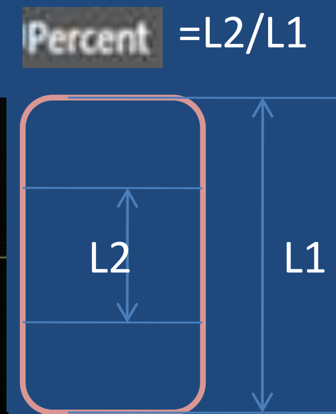


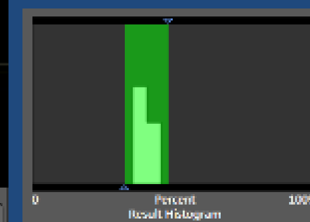
Effect for two direction choice



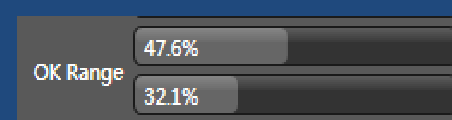
Effect for two tracking mode







Search a light source as top of image , select Direction choice and tracking mode which you need ,set ok range， click on inspect icon， it will automatically calculate the percentage result ， packing the range of the result histogram if it is regular distribution



If wrong part , the percentage result is 25.08% less than 32.1%, it will alarm